

Week1: Biochemistry



Toothpaste For Dinner.com

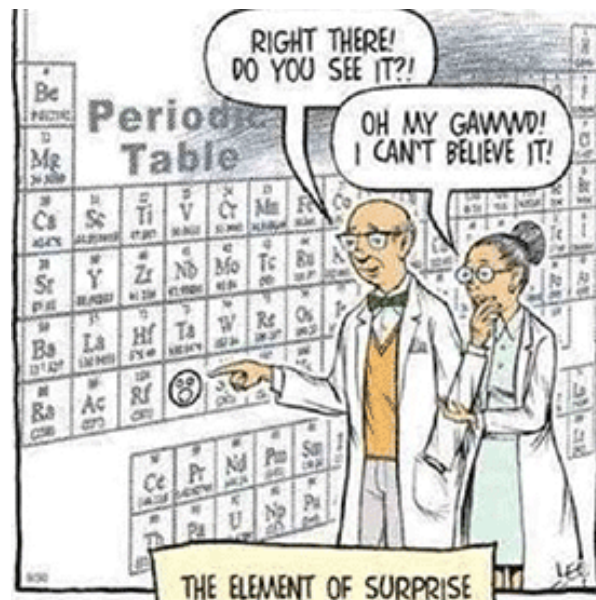
Campbell Chapters

2: [Atoms & Molecules](#)

3: [The Chemistry of Water](#)

4: [Carbon: The Basis of Molecular Diversity](#)

5: [Biological Macromolecules & Lipids](#)



A revision....

ATOMS AND MOLECULES

Matter

- Takes up space and has **mass**
- Exists as **elements** (pure form) and in chemical combinations called **compounds**

STATES OF MATTER



Elements

- Can't be broken down into simpler substances by chemical reaction
- Composed of **atoms**

Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	*	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	**	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo
			*	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
			**	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

The Elements of Life

- Essential elements in living things include carbon C, hydrogen H, oxygen O, and nitrogen N making up 96% of an organism
- A few other elements make up the remaining 4% of living matter

Table 2.1 Elements in the Human Body

Element	Symbol	Percentage of Body Mass (including water)	
Oxygen	O	65.0%	} 96.3%
Carbon	C	18.5%	
Hydrogen	H	9.5%	
Nitrogen	N	3.3%	
Calcium	Ca	1.5%	} 3.7%
Phosphorus	P	1.0%	
Potassium	K	0.4%	
Sulfur	S	0.3%	
Sodium	Na	0.2%	
Chlorine	Cl	0.2%	
Magnesium	Mg	0.1%	
Trace elements (less than 0.01% of mass): Boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), iron (Fe), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), zinc (Zn)			

Trace Elements

- Trace elements are required by an organism in only minute quantities
- Minerals such as **Fe** and **Zn** are trace elements



Deficiencies

- If there is a deficiency of an essential element, disease results



(a) Nitrogen deficiency



**(b) Iodine deficiency
(Goiter)**

Compounds

- Are substances consisting of **two or more elements** combined in a **fixed ratio**
- Have characteristics different from those of their elements



Na
Sodium

+



+

Cl
Chlorine

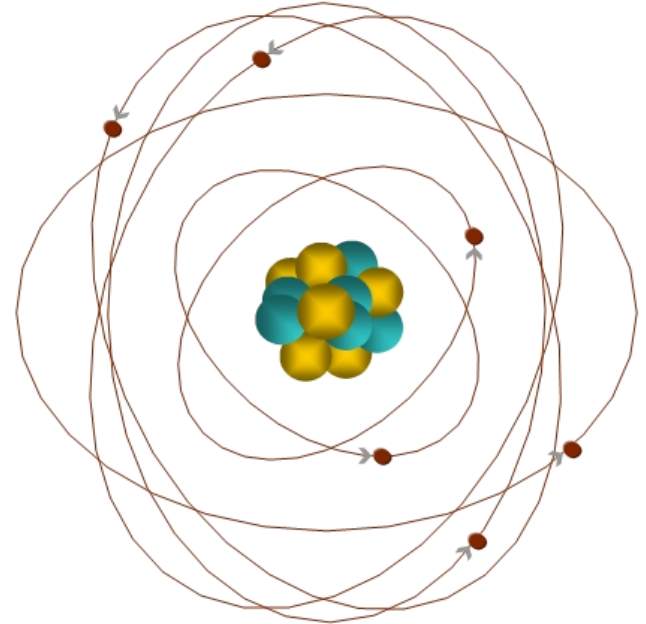
→



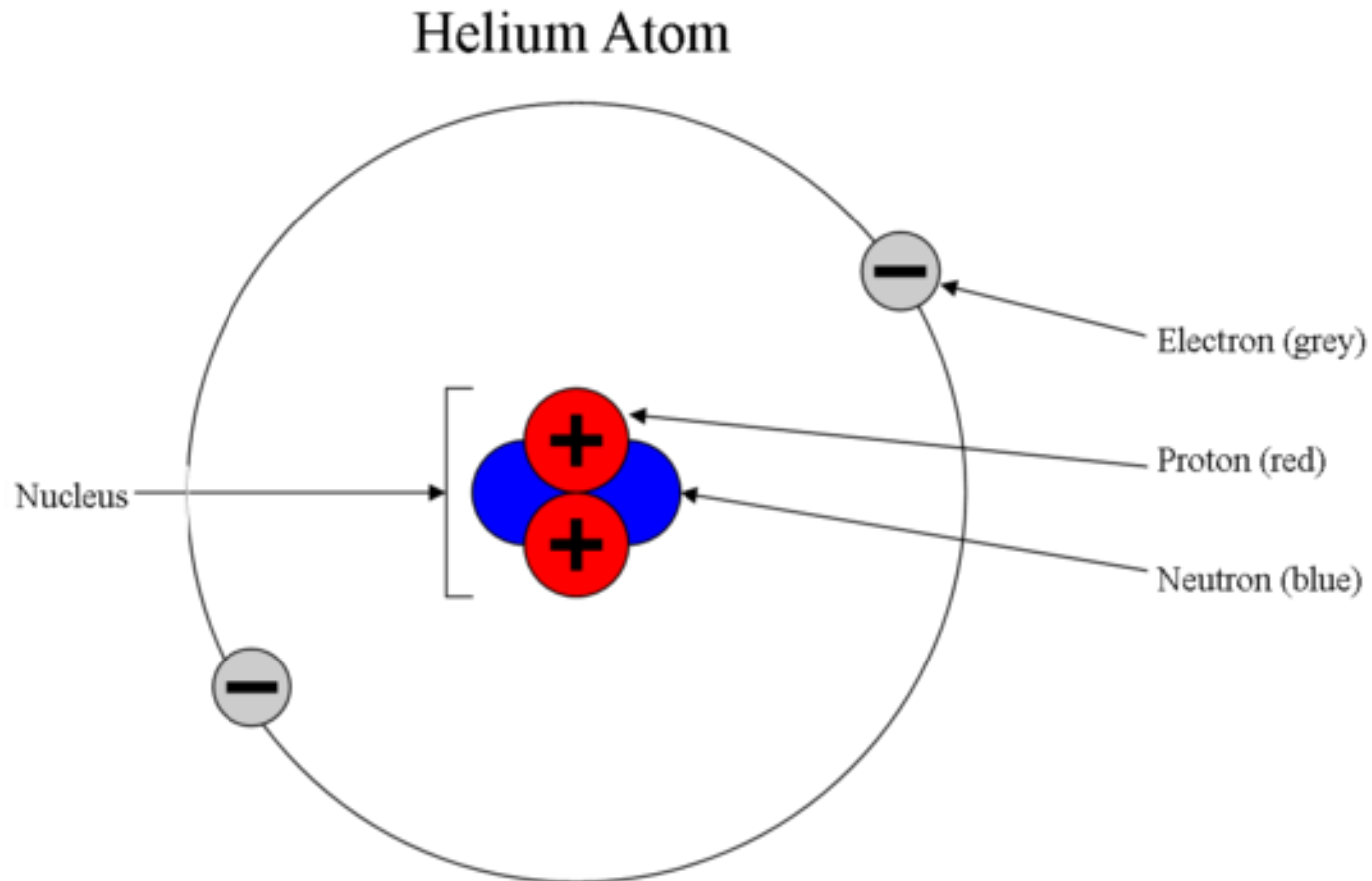
NaCl
Sodium chloride

Properties of Matter

- An element's properties depend on the structure of its atoms
- Each element consists of a certain kind of atom that is different from those of other elements
- An atom is the smallest unit of matter that still retains the properties of an element

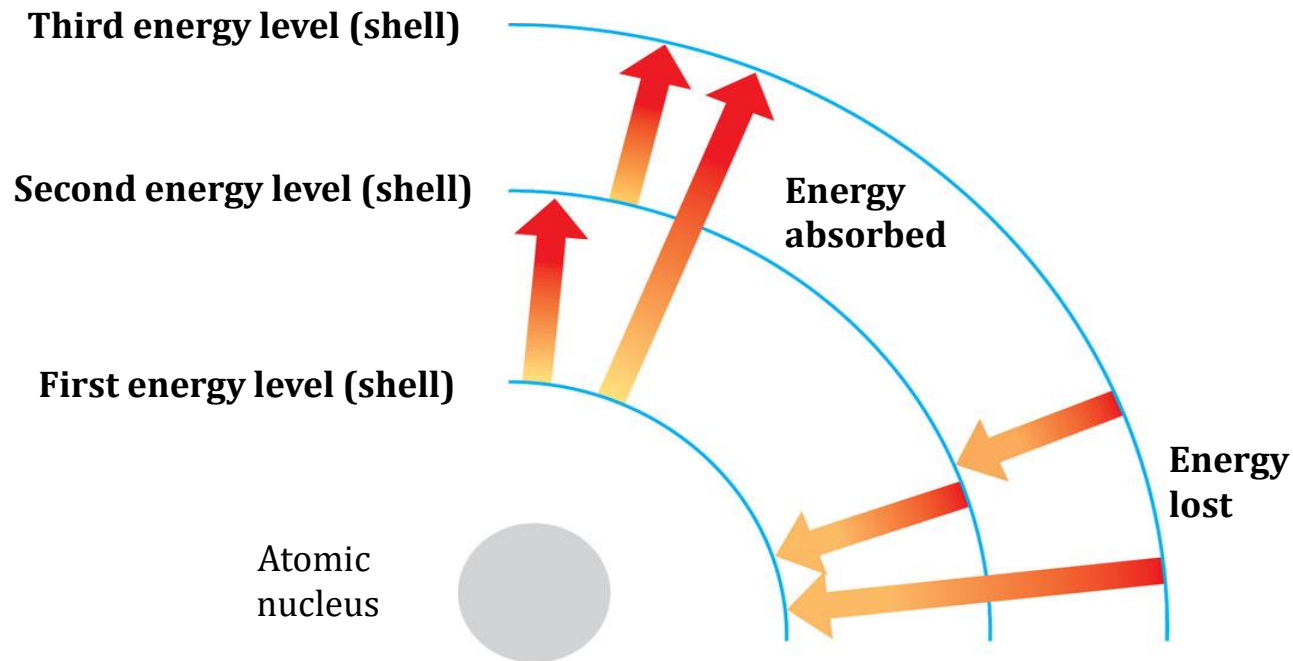


Sub atomic particles:



Energy Levels

- Are represented by electron shells

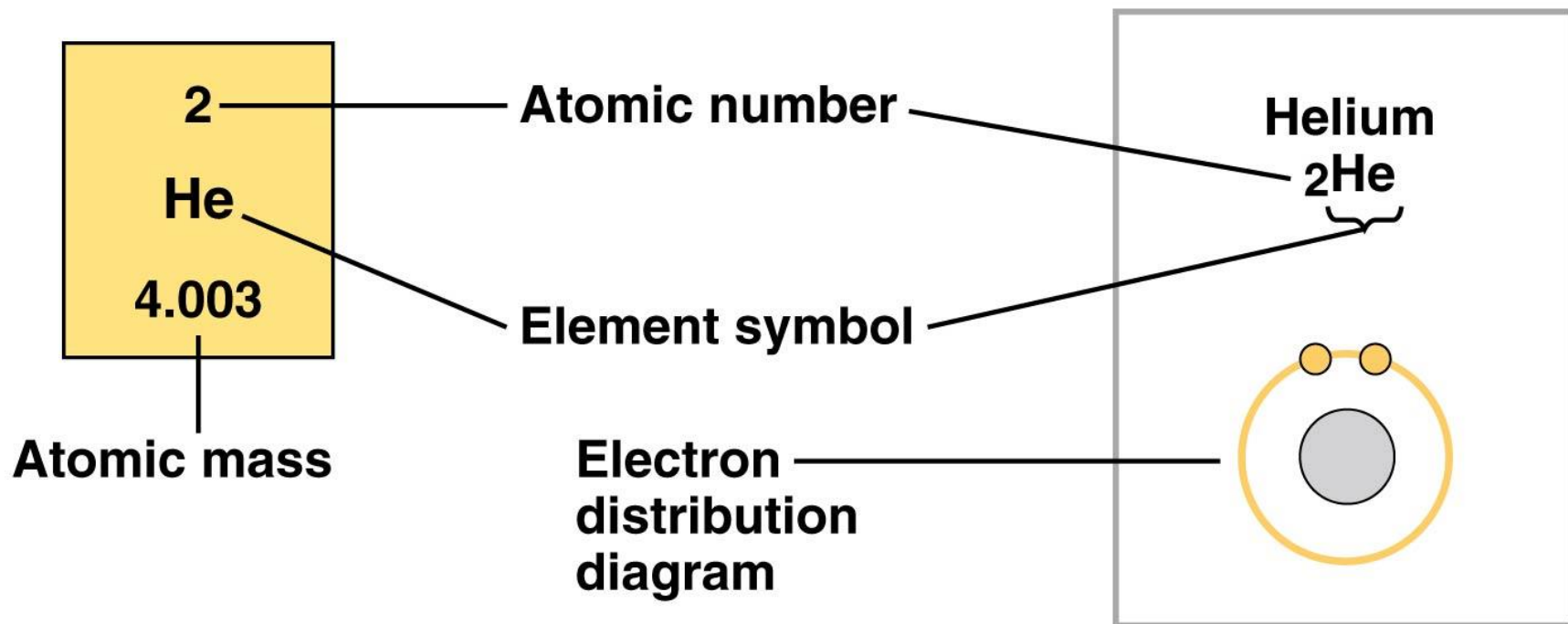


- (b) An electron can move from one level to another only if the energy it gains or loses is exactly equal to the difference in energy between the two levels. Arrows indicate some of the step-wise changes in potential energy that are possible.

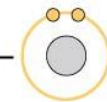
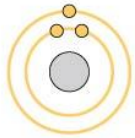
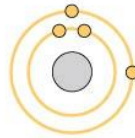
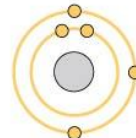
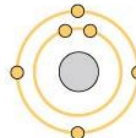
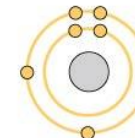
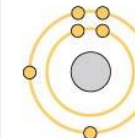
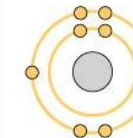
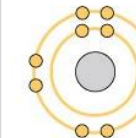
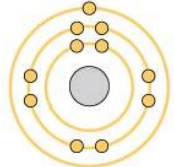
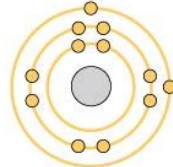
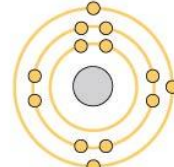
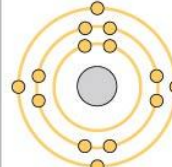
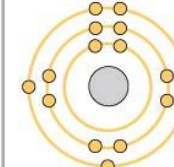
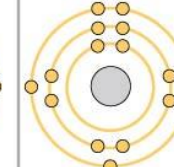
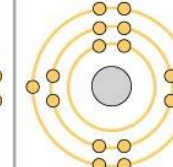
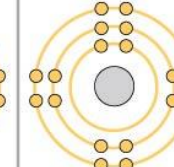
Atomic Number & Atomic Mass

- Atoms of the various elements differ in their number of subatomic particles
- The **number of protons** in the nucleus = **atomic number**
- The number of **protons** + **neutrons** = **atomic mass**
- Neutral atoms have equal numbers of protons & electrons (+ and – charges)

Electron distribution diagram

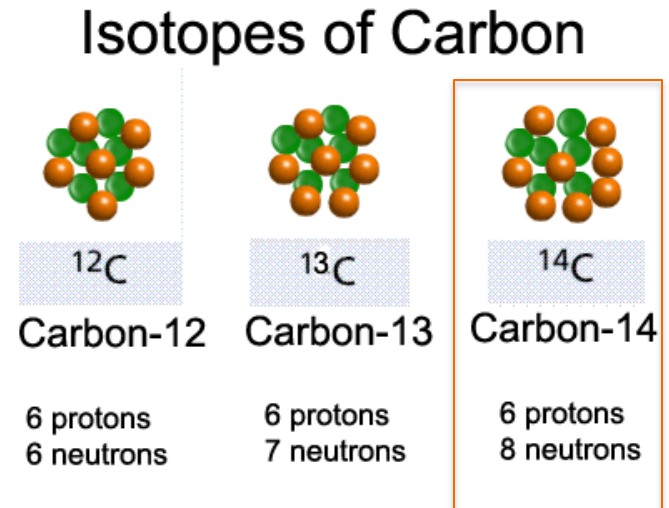
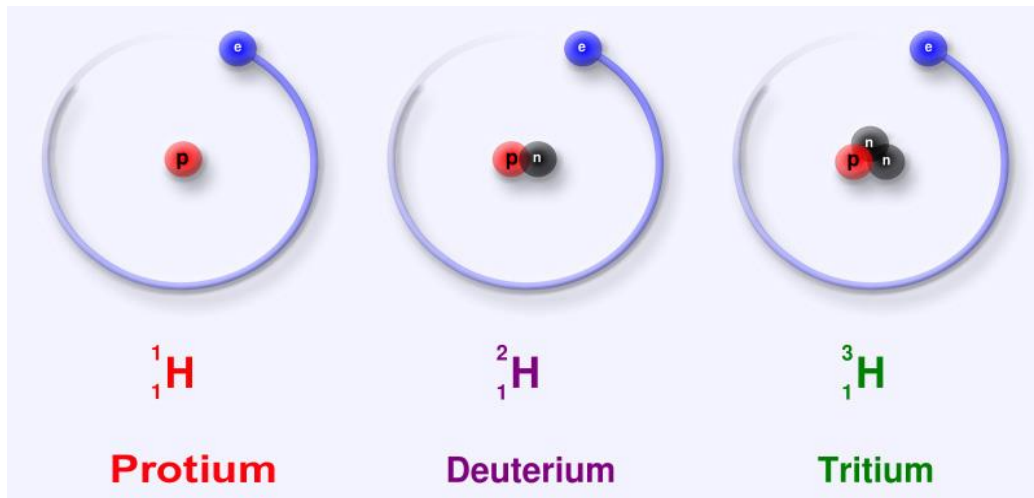


Electron distribution diagram for the first 18 elements

First shell	Hydrogen ${}^1_1\text{H}$ 							2 Atomic number He Element symbol 4.003 Atomic mass Electron distribution diagram	Helium ${}^2_2\text{He}$ 
Second shell	Lithium ${}^3_3\text{Li}$ 	Beryllium ${}^4_4\text{Be}$ 	Boron ${}^5_5\text{B}$ 	Carbon ${}^6_6\text{C}$ 	Nitrogen ${}^7_7\text{N}$ 	Oxygen ${}^8_8\text{O}$ 	Fluorine ${}^9_9\text{F}$ 	Neon ${}^{10}_{10}\text{Ne}$ 	
Third shell	Sodium ${}^{11}_{11}\text{Na}$ 	Magnesium ${}^{12}_{12}\text{Mg}$ 	Aluminum ${}^{13}_{13}\text{Al}$ 	Silicon ${}^{14}_{14}\text{Si}$ 	Phosphorus ${}^{15}_{15}\text{P}$ 	Sulfur ${}^{16}_{16}\text{S}$ 	Chlorine ${}^{17}_{17}\text{Cl}$ 	Argon ${}^{18}_{18}\text{Ar}$ 	

Isotopes

- Same number of protons, different number of neutrons



Isotopes

- May be **radioactive**, spontaneously giving off particles and energy → radioactive decay
- May be used to date fossils (C-14) or as medical tracers (I-131)

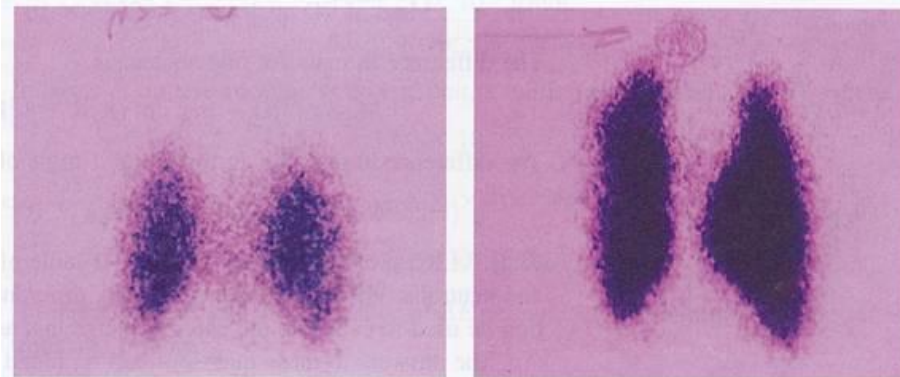


FIGURE 18.8
After consumption of Na^{131}I , the patient's thyroid is scanned for radioactivity levels to determine the efficiency of iodine absorption. (left) A normal thyroid. (right) An enlarged thyroid.

CHEMICAL BONDS

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graph TD; A[CHEMICAL BONDS] --> B[Covalent]; A --> C[Ionic]; A --> D[Hydrogen]; A --> E[Van der Waals]; B --> F[Polar]; B --> G[Non-polar];
```

Covalent

Ionic

Hydrogen

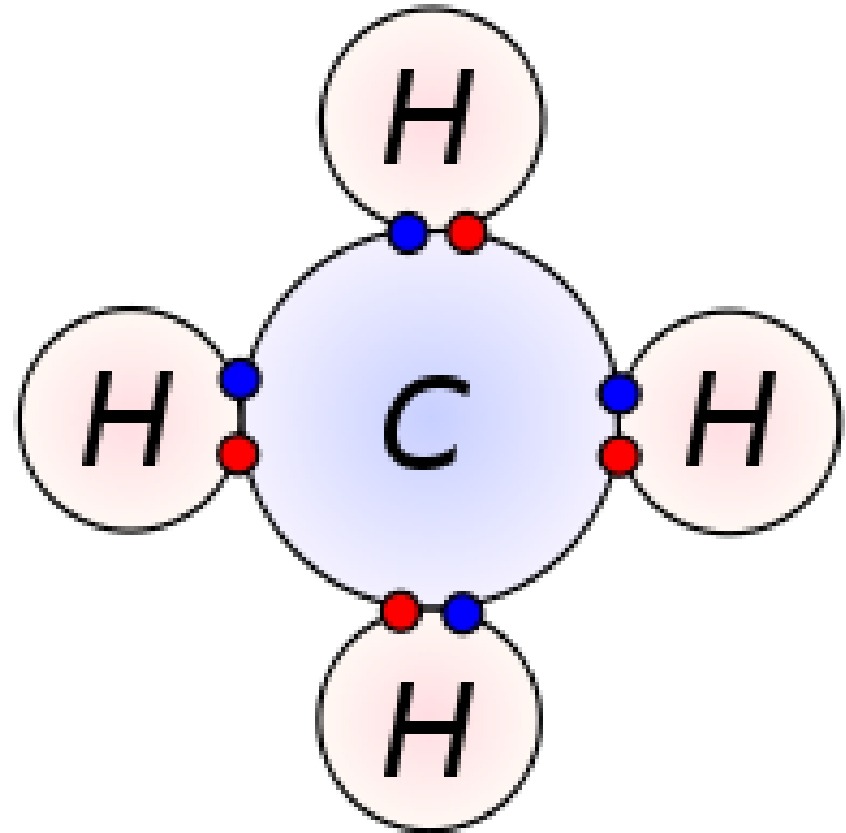
**Van der
Waals**

Polar

Non-polar

Covalent Bonds

- Sharing of at least a pair of **valence electrons**
 - Electrons in the outermost shell
- Examples: CH_4 , H_2 , O_2



● Electron from hydrogen

● Electron from carbon

Covalent Bonding in 4 molecules

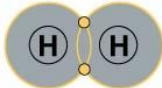
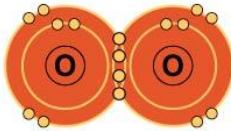
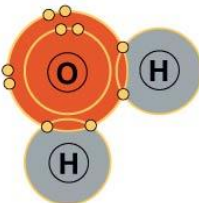
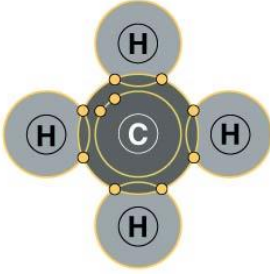
Fill in the Blanks

___hydrogen atoms share ___
pair of electrons, forming a
___ bond

___oxygen atoms share ___
pair of electrons, forming a
___ bond

___hydrogen atoms and ___
oxygen atom(s) are joined
by ___ bonds, forming a
_____ of water

___hydrogen atoms can
satisfy the valence of one
_____ atom, forming
methane.

Name and Molecular Formula	Electron Distribution Diagram	Lewis Dot Structure and Structural Formula	Space-Filling Model
(a) Hydrogen (H ₂)		H:H H—H	
(b) Oxygen (O ₂)		Ö::Ö O=O	
(c) Water (H ₂ O)		:Ö:H H O—H H	
(d) Methane (CH ₄)		H H:C:H H H H—C—H H	

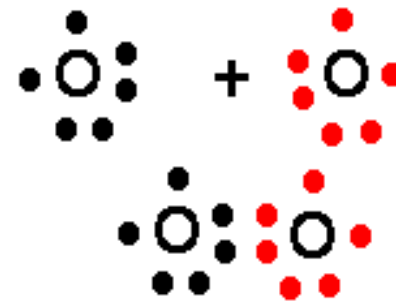
Covalent Bonding

- **Electronegativity**
 - Is the attraction of a particular kind of atom for the electrons in a covalent bond
- The more electronegative an atom
 - The more strongly it pulls shared electrons toward itself

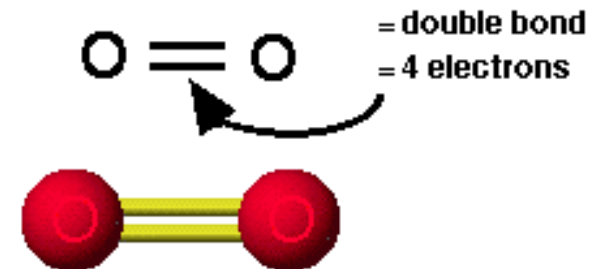
Covalent Bonding

- In a **nonpolar covalent bond**
 - The atoms have **similar electronegativities**
 - Share the electron equally

Covalent Bonding -
Oxygen Molecule, O_2



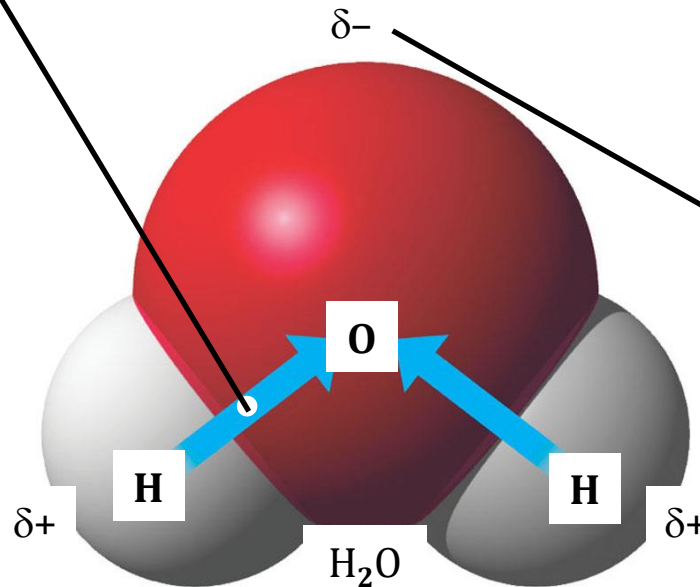
Equal Sharing of electrons between two identical non-metals.



Covalent Bonding

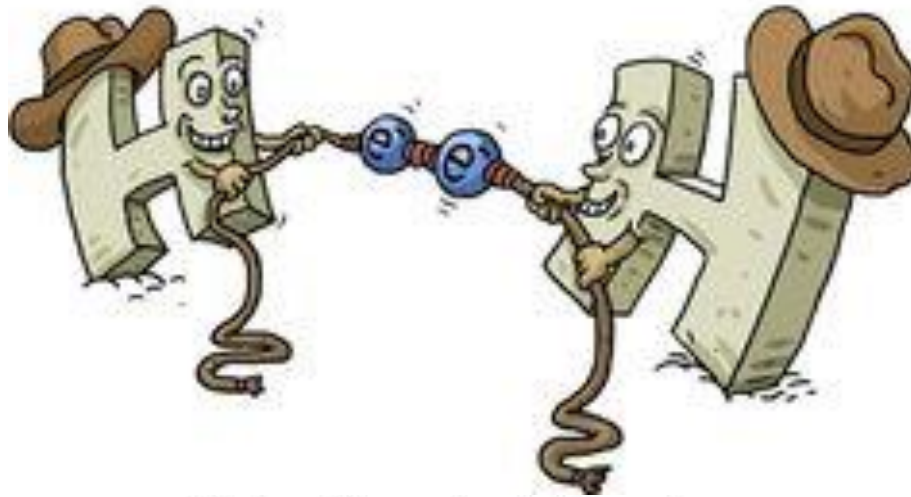
- In a **polar covalent bond**
 - The atoms have **different electronegativities**
 - Share the electrons **unequally**

Because oxygen (O) is more electronegative than hydrogen (H),
shared electrons are pulled more toward oxygen.



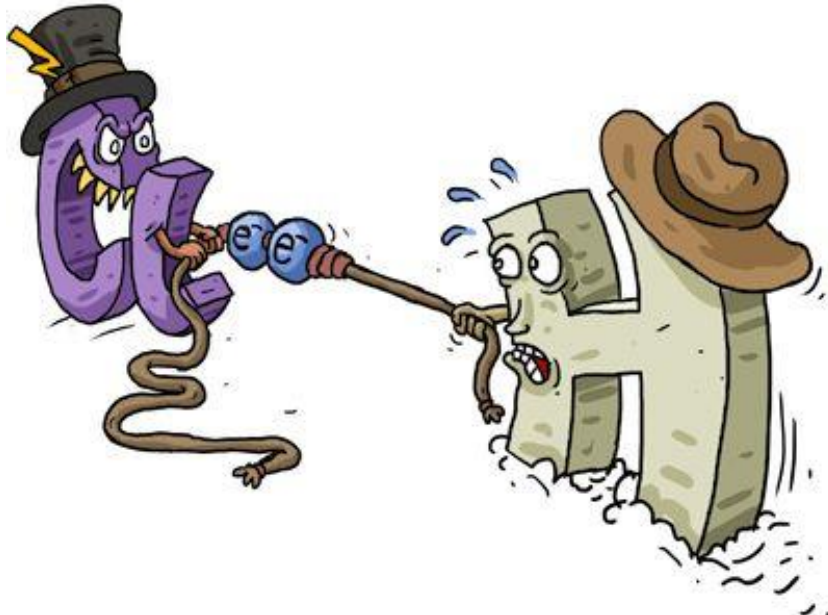
This results in a partial negative charge on the oxygen and a partial positive charge on the hydrogens.

Non-Polar Covalent Bond



- Balanced
- Symmetrical

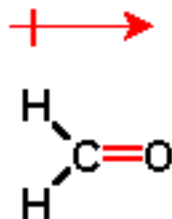
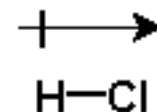
Polar Covalent Bond



- Unbalanced
- asymmetrical

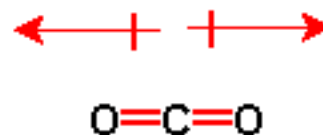
Polar bond vs. Polar molecule

- Bond polarity
 - Difference in electronegativities
- Molecule polarity
 - Sum of all bond polarities in the molecule



$$\mu = 2.2 \text{ D}$$

polar



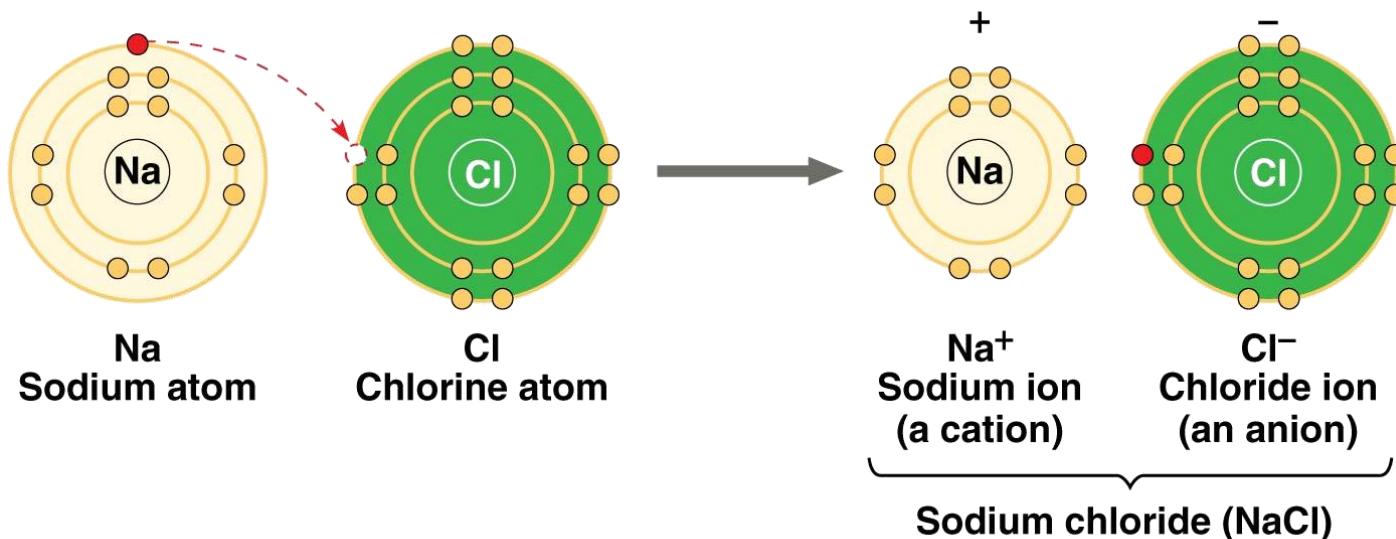
$$\mu = 0 \text{ D}$$

nonpolar

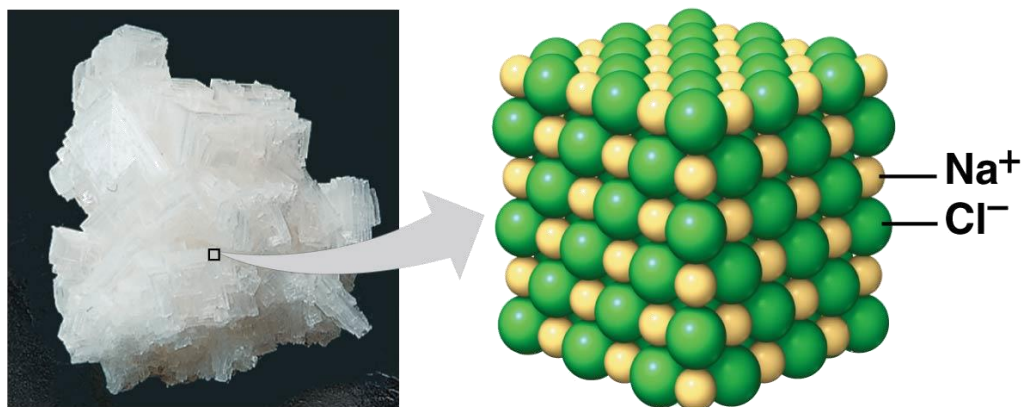
Ionic Bonds

- Electron transfer between two atoms
 - Anion (-ve), Cation (+ve)
- Metals+Non-metals = ionic compound
 - Ionic compounds are often called salts, which may form crystals

Electron transfer and ionic bonding



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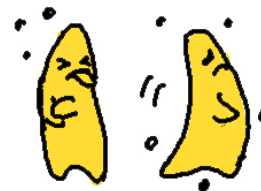
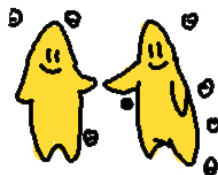
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Covalent bonding

Bobby was always a strange little nonmetal... he never liked to bond covalently. Sometimes, just to give it a try, he'd bond with another nonmetal....

And he'd be happy for a while, because he'd have filled his octet.

But no, he wasn't really one who wanted to SHARE all the time... And after a period, he'd stop.



Ionic bonding

Miss, you dropped an electron....

Oh, why thank you young man!!!



Where they'd be attracted and bonded by charge and not through sharing... ooh ooh ooh!!!! :D

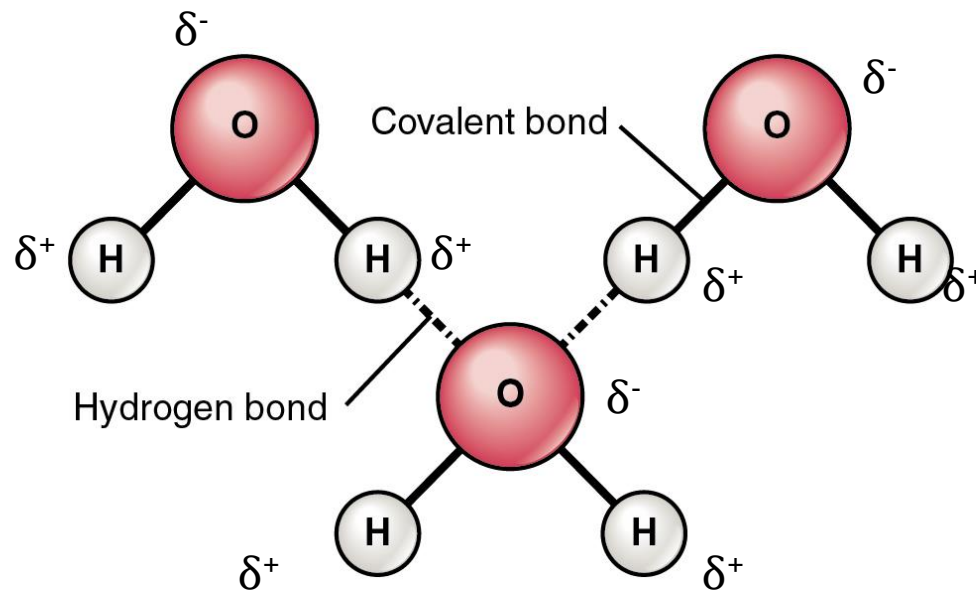


And it was so that he met his life's love, Millie the Metalloid :D

No, he was always more into ionic bonding....

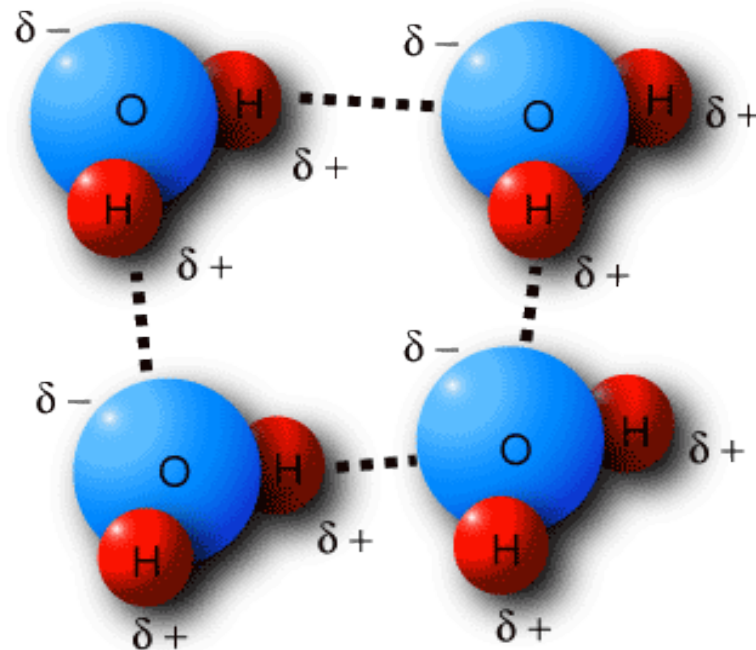
Hydrogen Bonds

- Between positively charged H atom and the strongly electronegative O /N/F of another molecule → intermolecular



Van der Waals Interactions

- Between molecules → intermolecular
- Occur when transiently positive and negative regions of molecules attract each other



ACTIVITY

Form as many groups in your class to role play compounds with different bonding.

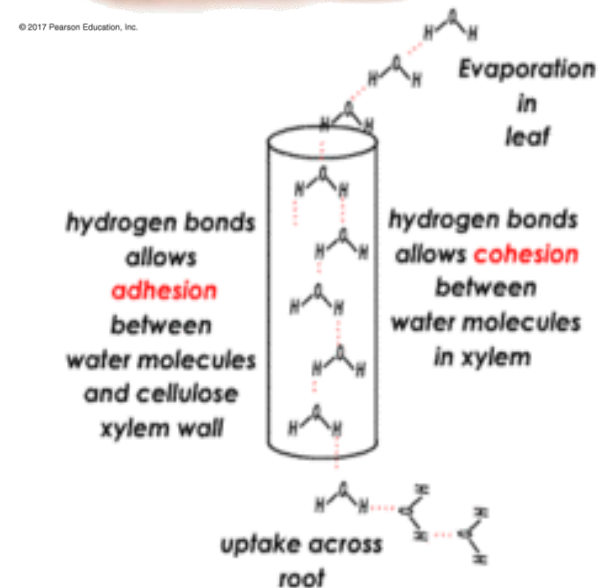
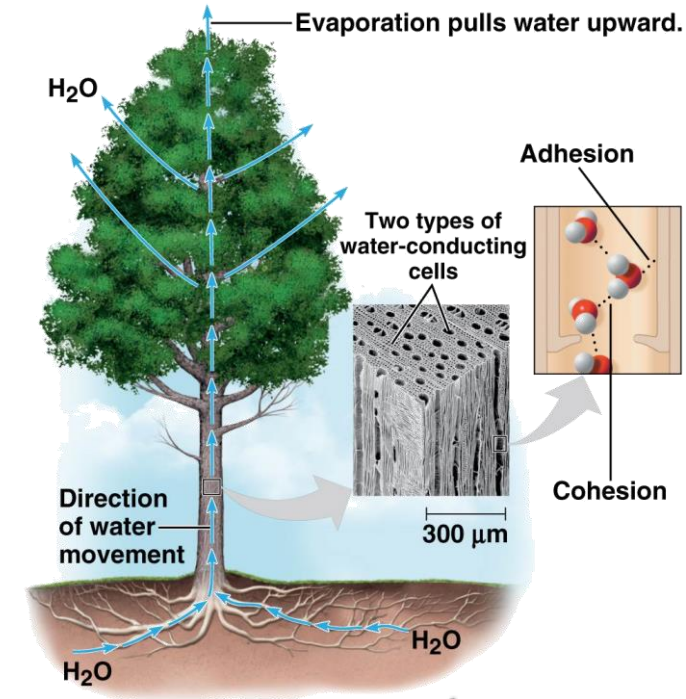




THE CHEMISTRY OF WATER

1. Cohesive behavior

- **Cohesion**: → hydrogen bonds
 - high surface tension, difficult to stretch/break surface of liquid
 - Help transport of water upwards in plants
- **Adhesion**: water adhere to *unlike* molecules
 - Water – plant cell-wall (hydrogen bond)
 - Help counter the downward pull of gravity
- **Surface tension**,
 - Related to cohesion
 - Ease to stretch/break surface of liquid



2. Moderation of Temperature

- High Specific Heat,

- the amount of energy needed to increase the temperature of 1 gram of a substance by 1°C (changes temp slowly) $\rightarrow 1 \text{ cal/gram}\cdot^{\circ}\text{C}$
- Water changes temperature less than other liquids
- Due to hydrogen bonding,
 - heat absorbed to break H-bonds
 - Heat released to form H-bonds
 - 1 cal heat $\rightarrow$ small Δ in water's temperature

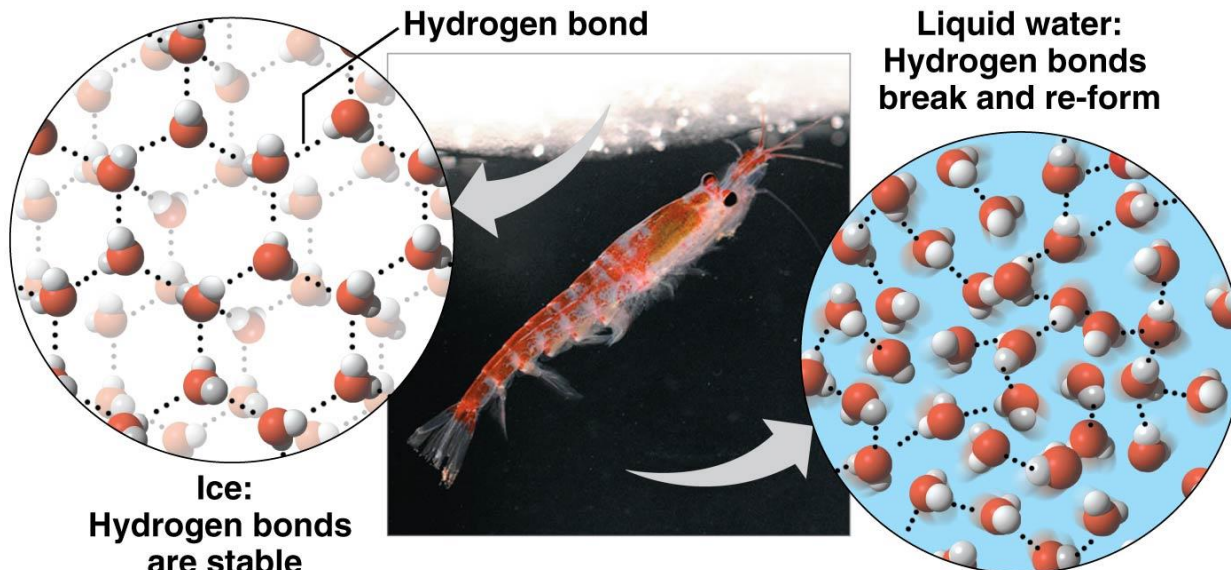
Water can moderate air temperature
in coastal areas



- **High Heat of Vaporization**, the amount of heat energy needed to change 1g liquid to a gas (or evaporate)
 - Hottest liquid molecules (greatest kinetic energy) leave surface as gas
 - Hydrogen bonds need to be broken
 - **Evaporative cooling**
 - Stability of temperature in lakes and ponds
 - Dissipation of body heat by sweating
- **High Heat of Fusion**, the energy needed to convert a substance from a liquid to a solid (freezing, in the case of water)

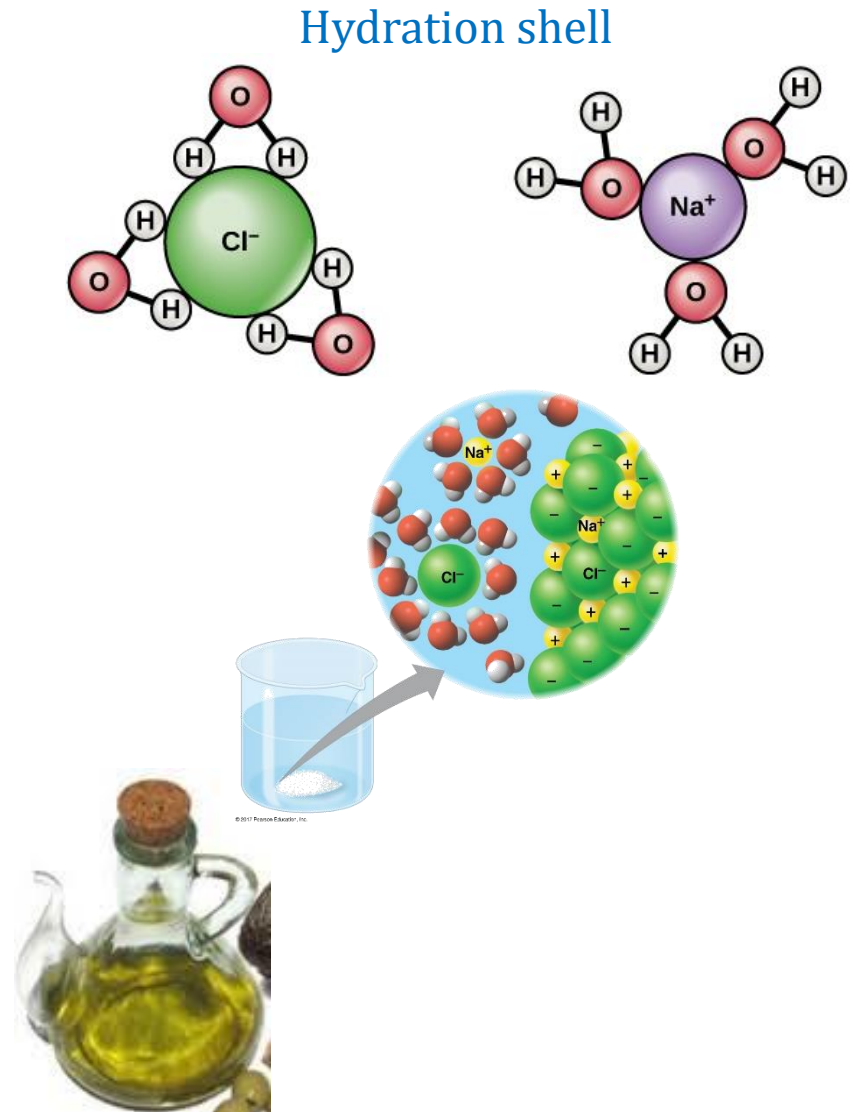
3. Floating of ice on liquid water

- $>4^{\circ}\text{C}$ $\rightarrow$ water is liquid
- $0-4^{\circ}\text{C}$ $\rightarrow$ water begins to freeze
- 0°C $\rightarrow$ H_2O molecules form stable H-bonds
- Solid water is less dense than liquid water, $\rightarrow$ ice floats.
- Allows life to exist under frozen surfaces of lakes and sea



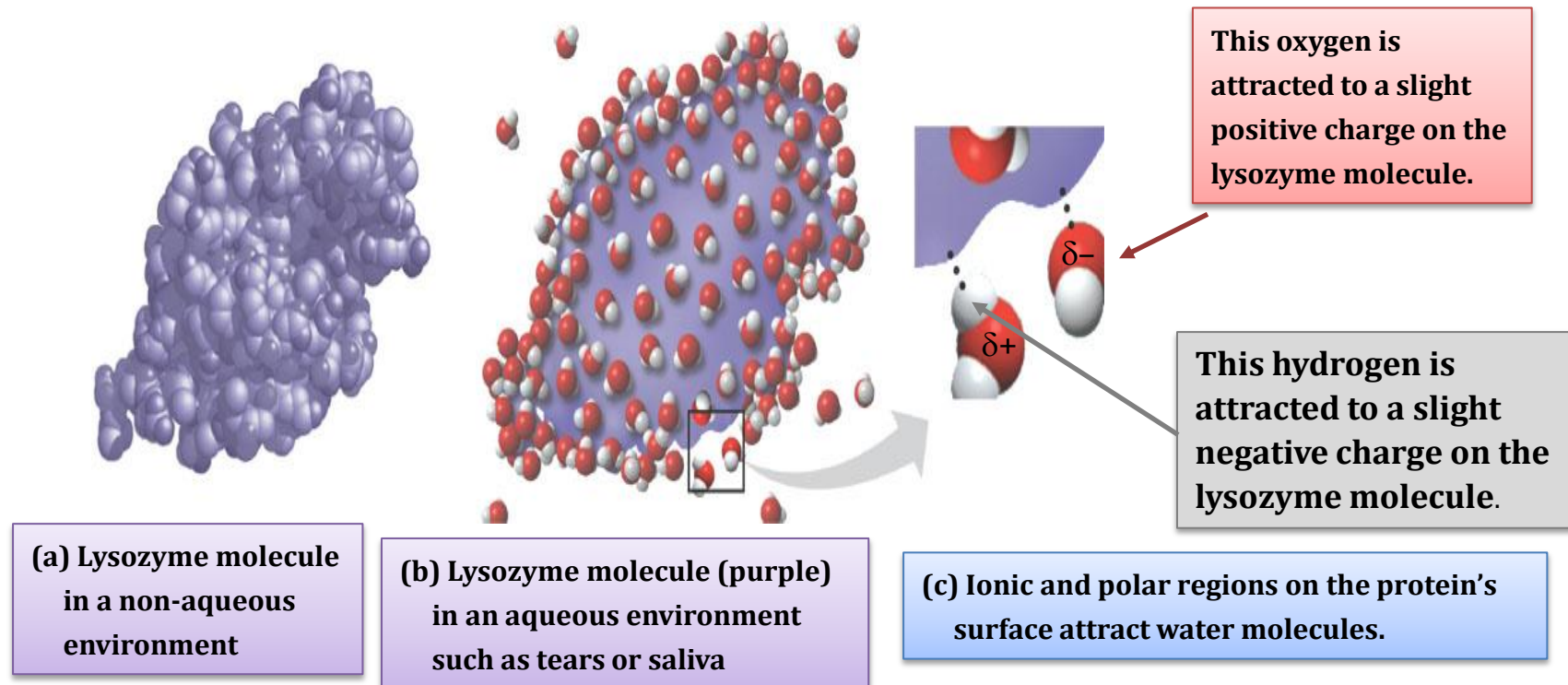
4. A universal solvent

- **As the universal solvent**, highly polar compound, dissolves polar substances and ionic compounds
 - Hydrophilic → ionic and polar
 - Hydrophobic → non-ionic and non-polar



Water as a solvent

Figure 3.9 *A water soluble protein*

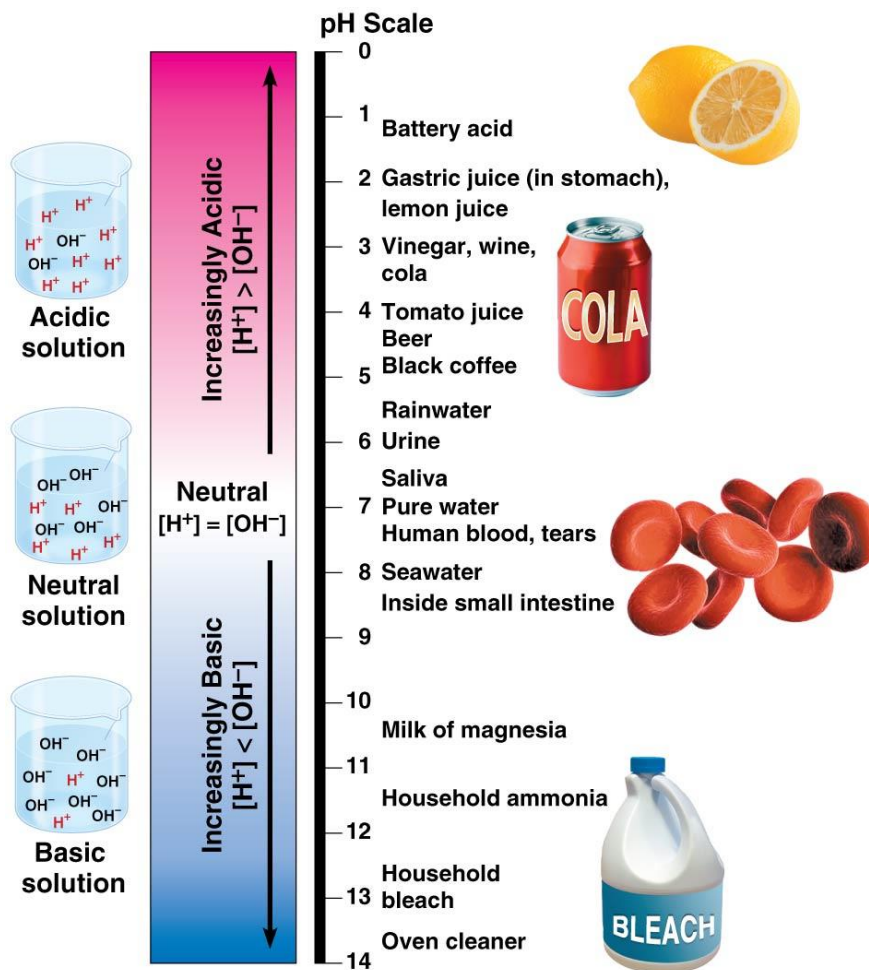


pH (power of Hydrogen)

- pH scale runs between 0 and 14 and measures the relative acidity and alkalinity of aqueous solutions
- The **pH** of a solution is defined by the negative logarithm of H^+ concentration, written as

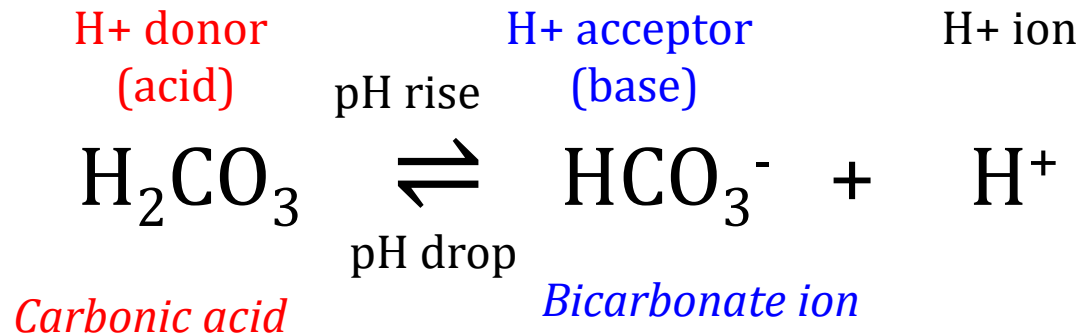
$$pH = -\log [H^+]$$

The pH scale and pH values of some aqueous solutions.



pH and buffers

- **Buffers:** Materials that have both acid and base properties
- Resist change in pH
- Absorb excess H^+ or donate H^+
- E.g. Bicarbonate ion HCO_3^-



Blood pH imbalance

SYMPTOMS OF ACIDOSIS

Central Nervous System

- Headache
- Sleepiness
- Confusion
- Loss of consciousness
- Coma

Respiratory System

- Shortness of breath
- Coughing

Heart

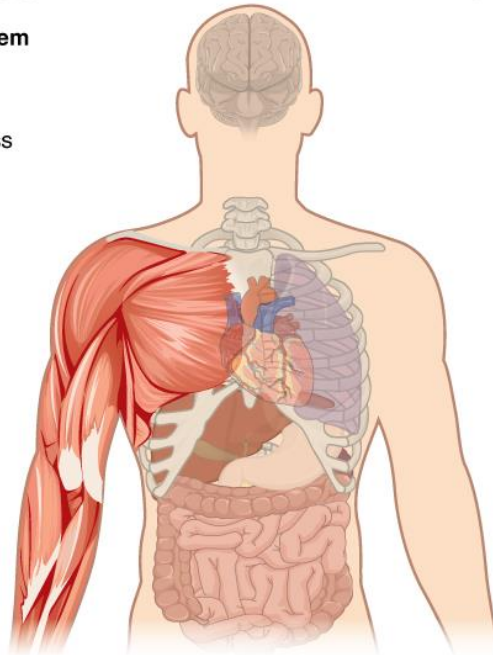
- Arrhythmia
- Increased heart rate

Muscular System

- Seizures
- Weakness

Digestive System

- Nausea
- Vomiting
- Diarrhea



SYMPTOMS OF ALKALOSIS

Central Nervous System

- Confusion
- Light-headedness
- Stupor
- Coma

Peripheral Nervous System

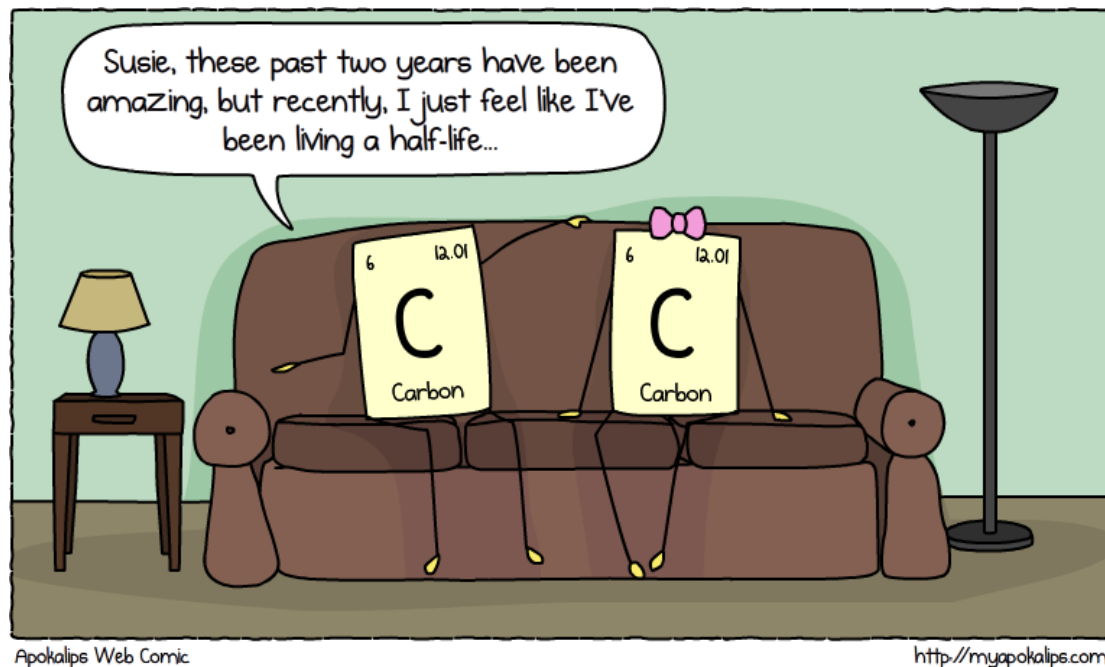
- Hand tremor
- Numbness or tingling in the face, hands, or feet

Muscular System

- Twitching
- Prolonged spasms

Digestive System

- Nausea
- Vomiting



CARBON CHEMISTRY

Carbon chemistry

- Carbon is the backbone of biological molecules (macromolecules)
- All living organisms are made up of chemicals based mostly on the element carbon

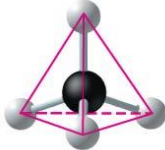
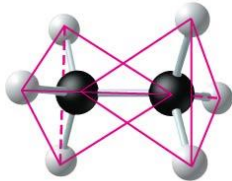
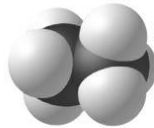
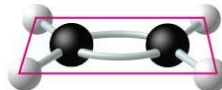


Carbon Chemistry

- **Organic chemistry** is the study of carbon compounds
- Carbon atoms can form diverse molecules by bonding to four other atoms
- Carbon compounds range from simple molecules to complex ones
- Carbon has four **valence electrons** and may form single, double, triple, or quadruple bonds

- The bonding versatility of carbon allows it to form many diverse molecules, including carbon skeletons

Figure 4.3 The shapes of three simple organic molecules.

Molecule	Molecular Formula	Structural Formula	Ball-and-Stick Model	Space-Filling Model
(a) Methane	CH ₄	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array} $		
(b) Ethane	C ₂ H ₆	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} $		
(c) Ethene (ethylene)	C ₂ H ₄	$ \begin{array}{c} \text{H} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array} $		

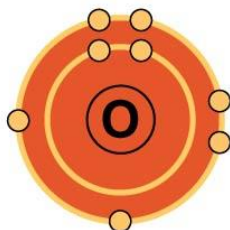
- The **electron configuration** of carbon gives it covalent compatibility with many different elements

Fig 4.4 Valences of the major elements of organic molecules.

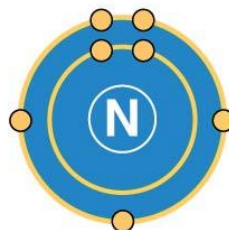
Hydrogen
(valence = 1)



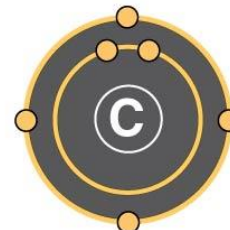
Oxygen
(valence = 2)



Nitrogen
(valence = 3)

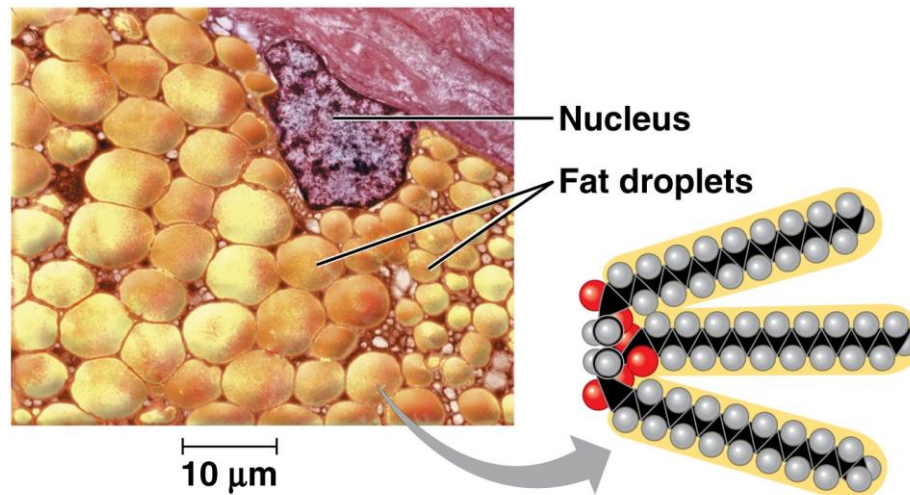


Carbon
(valence = 4)



Hydrocarbons

- Hydrocarbons are molecules consisting of **only carbon and hydrogen**
- Hydrocarbons are found in many of a cell's organic molecules



(a) Part of a human adipose cell

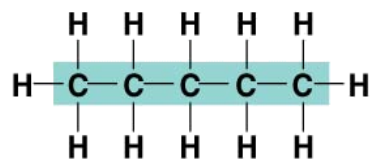
(b) A fat molecule

Isomers

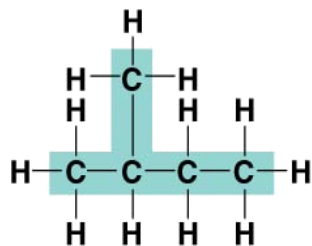
- Isomers are molecules with the **same molecular formula** but **different structures** and properties
- Three types of isomers are
 - Structural
 - Variation in covalent arrangement
 - Geometric/*cis-trans*
 - Variation in arrangement about a double bond
 - Enantiomers/optical isomers
 - Variation in spatial arrangement around an asymmetric carbon
 - Molecules are mirror image

3 types of isomers

(a) Structural isomers

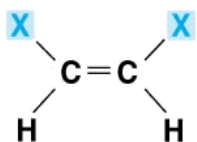


Pentane

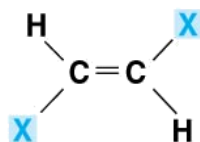


2-Methylbutane

(b) *Cis-trans* isomers

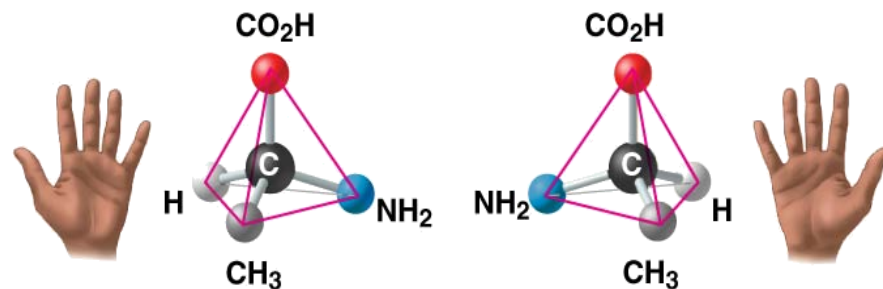


***cis* isomer:**
The two Xs are on the same side.



***trans* isomer:**
The two Xs are on opposite sides.

(c) Enantiomers



L isomer

D isomer

- Enantiomers are important in the pharmaceutical industry

Drug	Effects	Effective Enantiomer	Ineffective Enantiomer
Ibuprofen	Reduces inflammation and pain	 S-Ibuprofen	 R-Ibuprofen
Albuterol	Relaxes bronchial (airway) muscles, improving airflow in asthma patients	 R-Albuterol	 S-Albuterol

Chemical Groups

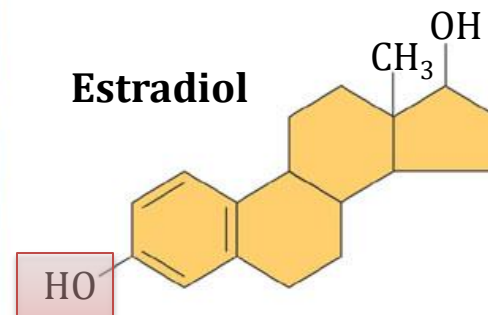
- Chemical groups that are directly involved in chemical reactions are called **functional groups**
- They are the **chemically reactive groups of atoms** within an organic molecule
- Give organic molecules distinctive **chemical properties**



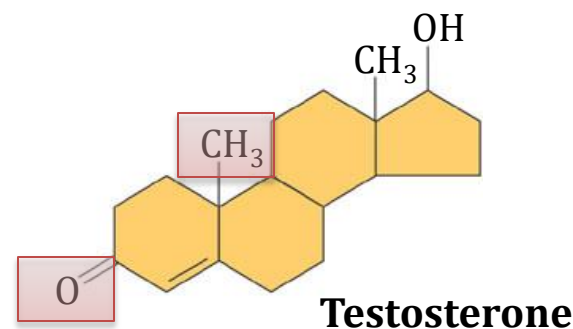
Female lion



Male lion



Estradiol

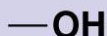
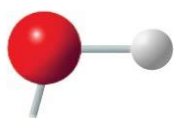


Testosterone

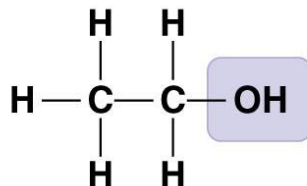
Figure 4.9

Chemical Groups

Hydroxyl group (—OH)



(may be written HO—)

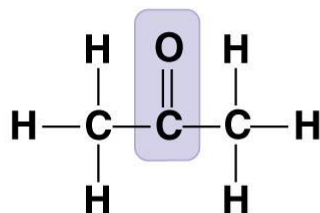
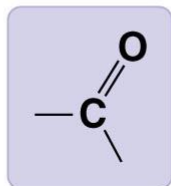
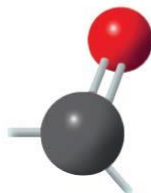


Ethanol, the alcohol
present in
alcoholic beverages

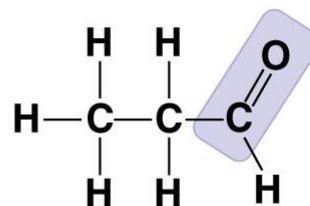
Polar due to electronegative oxygen. Forms hydrogen bonds with water.
Compound name: Alcohol

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Carbonyl group (>C=O)



Acetone,
the simplest ketone



Propanal,
an aldehyde

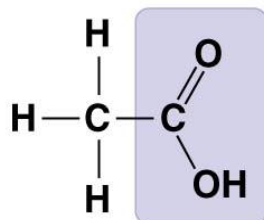
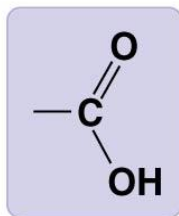
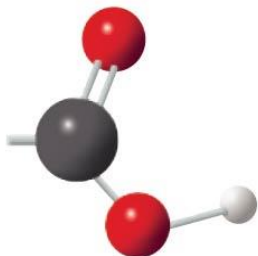
Sugars with ketone groups are called ketoses; those with aldehydes are called aldoses.

Compound name: Ketone or aldehyde

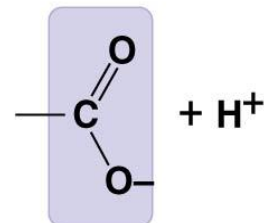
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Chemical Groups

Carboxyl group (—COOH)



Acetic acid, which gives vinegar its sour taste



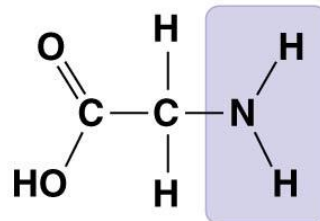
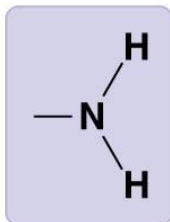
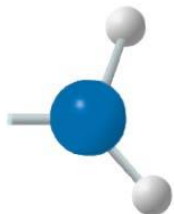
Ionized form of —COOH (carboxylate ion), found in cells

Acts as an acid.

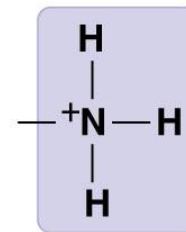
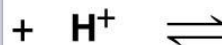
Compound name: Carboxylic acid, or organic acid

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Amino group (—NH₂)



Glycine, an amino acid (note its carboxyl group)



Ionized form of —NH₂, found in cells

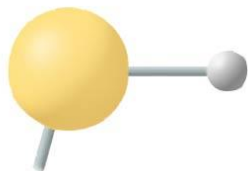
Acts as a base.

Compound name: Amine

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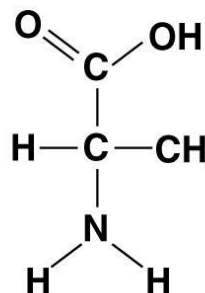
Chemical Groups

Sulfhydryl group (—SH)



—SH

(may be written HS —)



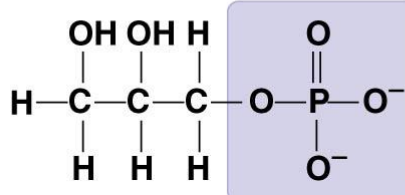
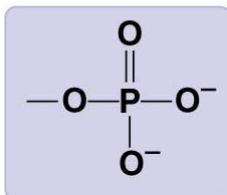
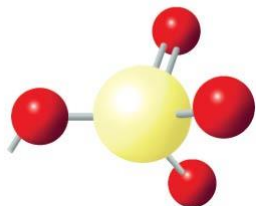
Cysteine, a sulfur-containing amino acid

Two —SH groups can react, forming a “cross-link” that helps stabilize protein structure.

Compound name: Thiol

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Phosphate group (—OPO_3^{2-})



Glycerol phosphate, which takes part in many important chemical reactions in cells

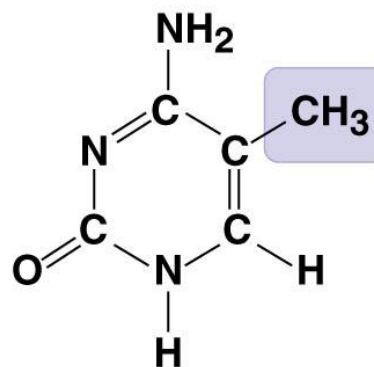
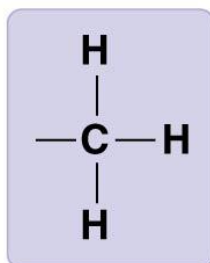
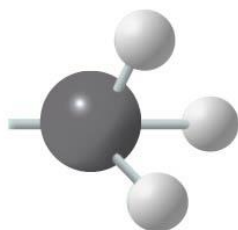
Contributes negative charge. When attached, confers on a molecule the ability to react with water, releasing energy.

Compound name: Organic phosphate

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Chemical Groups

Methyl group (—CH₃)



5-Methylcytosine, a component of DNA that has been modified by addition of a methyl group

Affects the expression of genes. Affects the shape and function of sex hormones.

Compound name: Methylated compound

BIOLOGICAL MACROMOLECULES & LIPIDS

Video/activity

- Biomolecules [Amoeba Sisters]
<https://www.youtube.com/watch?v=Y0244P1e9QM>

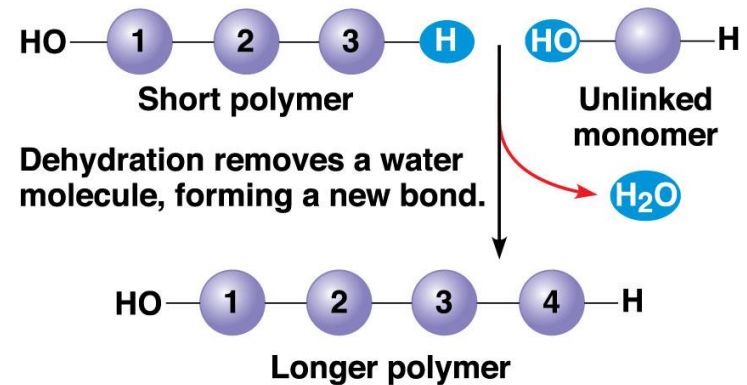
Organic Compounds

- Polymer: most macromolecules are polymers...a single unit repeated many times, hence “poly”mer
- Monomer: a single unit (“mono”mer) that makes up a polymer
- Macromolecule: a large organic molecule, made of multiple polymers
- Four classes of macromolecules (biomolecules)
 - **Carbohydrates**
 - **Lipids**
 - **Proteins**
 - **Nucleic Acids**

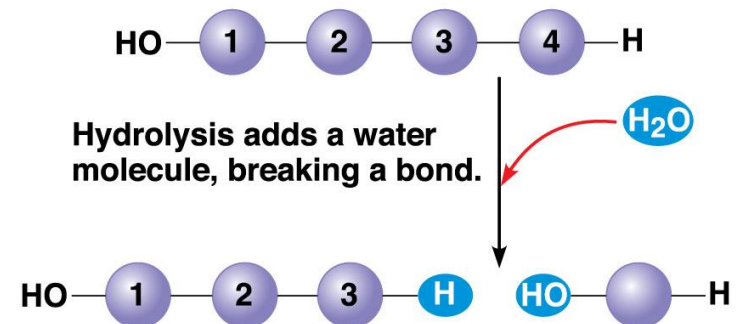
The Synthesis and Breakdown of Polymers

- Monomers form larger molecules by
 - **dehydration reaction/synthesis**
 - - H_2O molecule
- Polymers can disassemble by
 - **Hydrolysis**
 - + H_2O molecule

(a) Dehydration reaction: synthesizing a polymer



(b) Hydrolysis: breaking down a polymer



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Carbohydrate

- C, H, O
- $(\text{CH}_2\text{O})_n$, where n is any number from 3 to 8
- Serve as fuel and building material
- 3 classes:
 - Monosaccharides
 - Disaccharides
 - polysaccharides

Monosaccharides

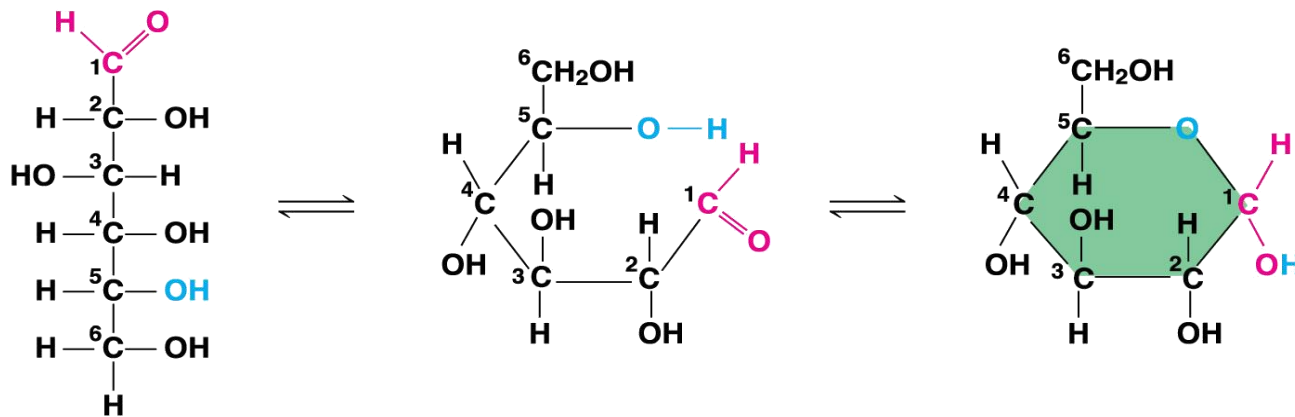
- Are the simplest sugars
- Can be used for fuel
- Can be converted into other organic molecules
- Can be combined into polymers
- glucose, galactose, fructose



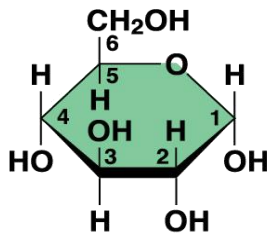
- Monosaccharides

- May be linear
- Can form rings

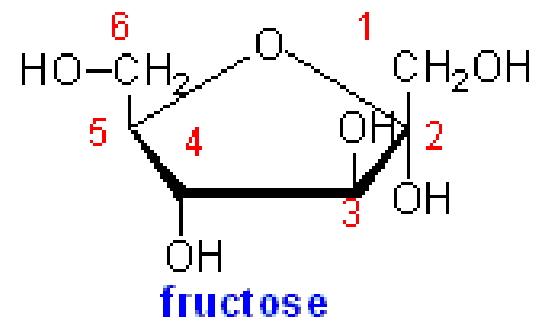
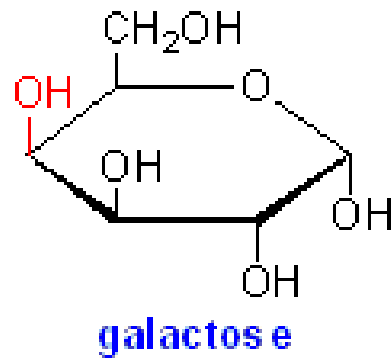
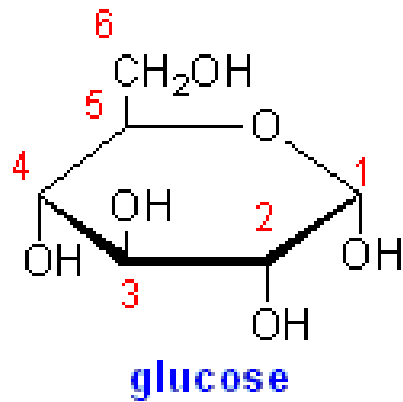
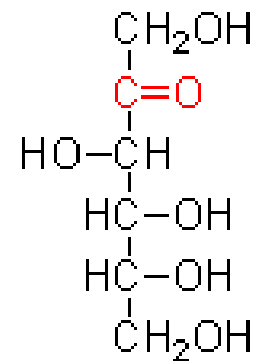
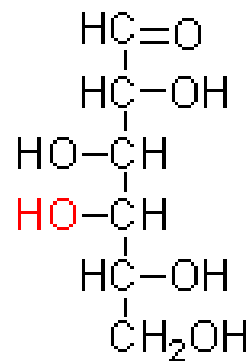
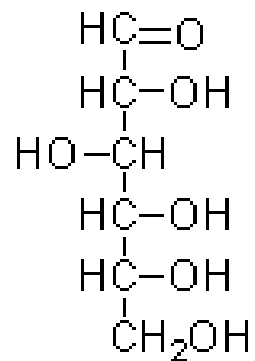
Chemical equilibrium between the linear and ring structures greatly favors the formation of rings



(a) Linear and ring forms

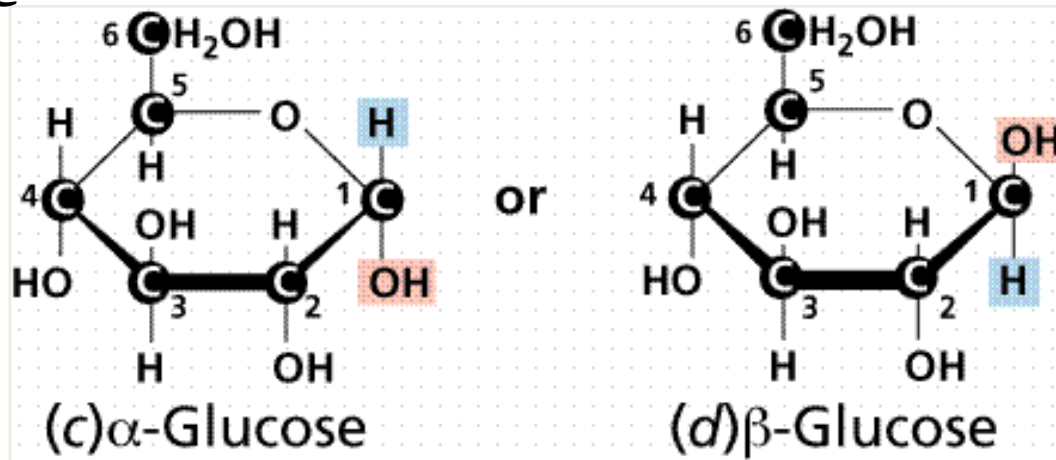


(b) Abbreviated ring structure

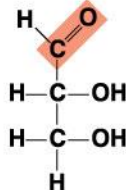
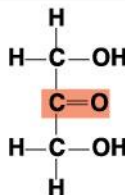
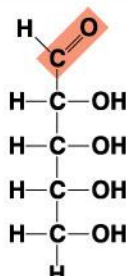
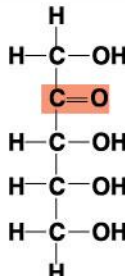
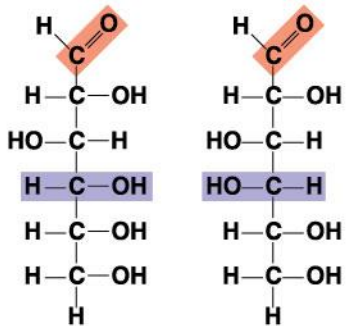
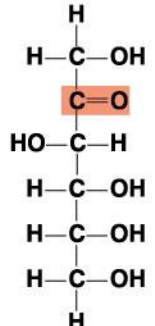


α - and β -glucose

- Two forms of glucose, seen below, differ only in a reversal of the H and OH on the first carbon.
- Even very small changes in the position of certain atoms may dramatically change the molecule



- Examples of monosaccharides
- Sugars vary in the position of the **carbonyl group**
 - Length of skeleton
 - The **chemical groups are arranged** around an asymmetric carbon

Aldoses (Aldehyde Sugars)	Ketoses (Ketone Sugars)
Trioses: three-carbon sugars (C ₃ H ₆ O ₃)	
	
Glyceraldehyde	Dihydroxyacetone
Pentoses: five-carbon sugars (C ₅ H ₁₀ O ₅)	
	
Ribose	Ribulose
Hexoses: six-carbon sugars (C ₆ H ₁₂ O ₆)	
	
Glucose Galactose	Fructose

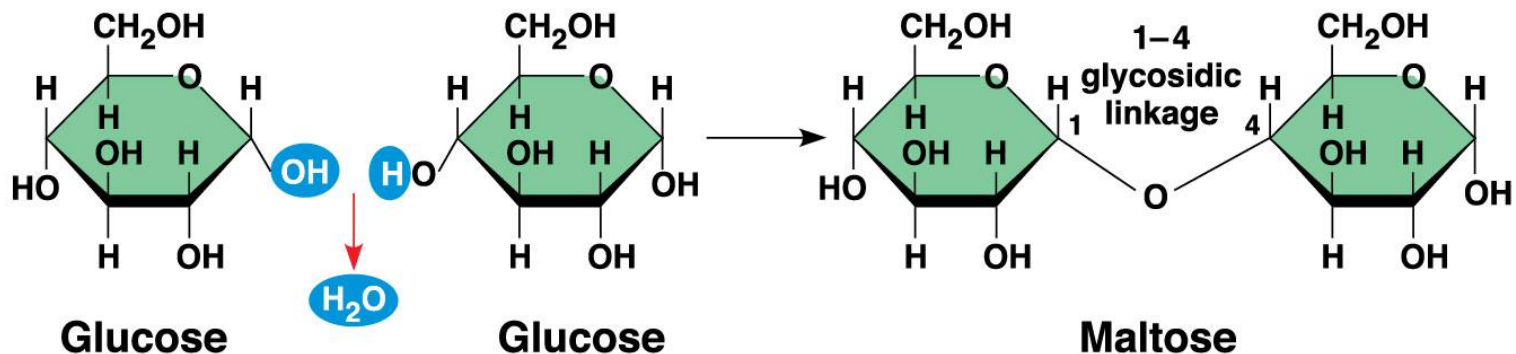
Disaccharides

- Di (two) sacchar (sugar): a double sugar that consists of two monosaccharides joined by a glycosidic linkage.
- Chemical formula – $C_{12}H_{22}O_{11}$
- **Glycosidic linkage:** covalent bond formed by a condensation/dehydration (water is lost) reaction **between two sugar monomers**

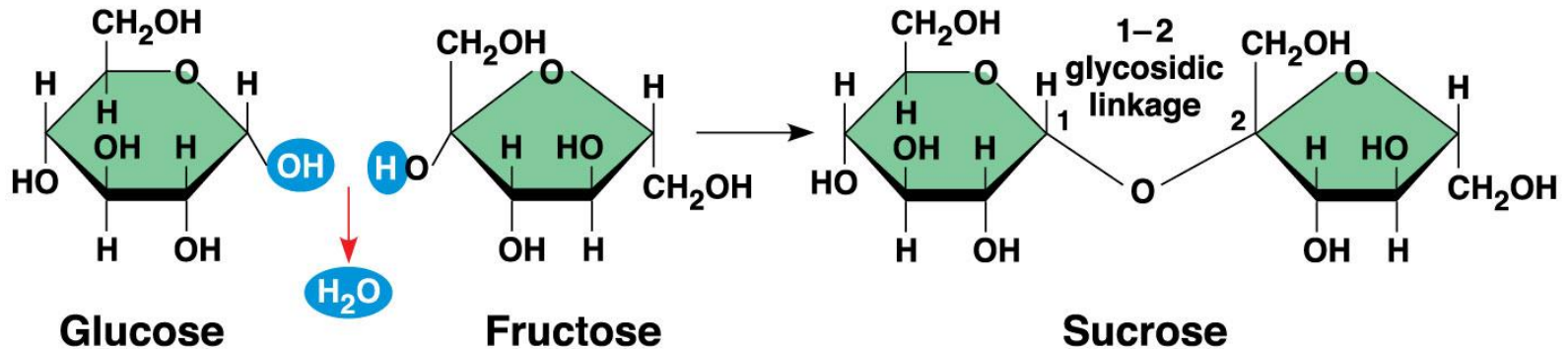
- Common Disaccharides:
 - Maltose= Glucose+Glucose (beer)
 - Lactose= Glucose+Galactose (milk)
 - Sucrose= Glucose+Fructose (table sugar/fruit sugar)
- All saccharides are storage molecules, they store energy to be used by living system

Disaccharide synthesis

(a) Dehydration reaction in the synthesis of maltose



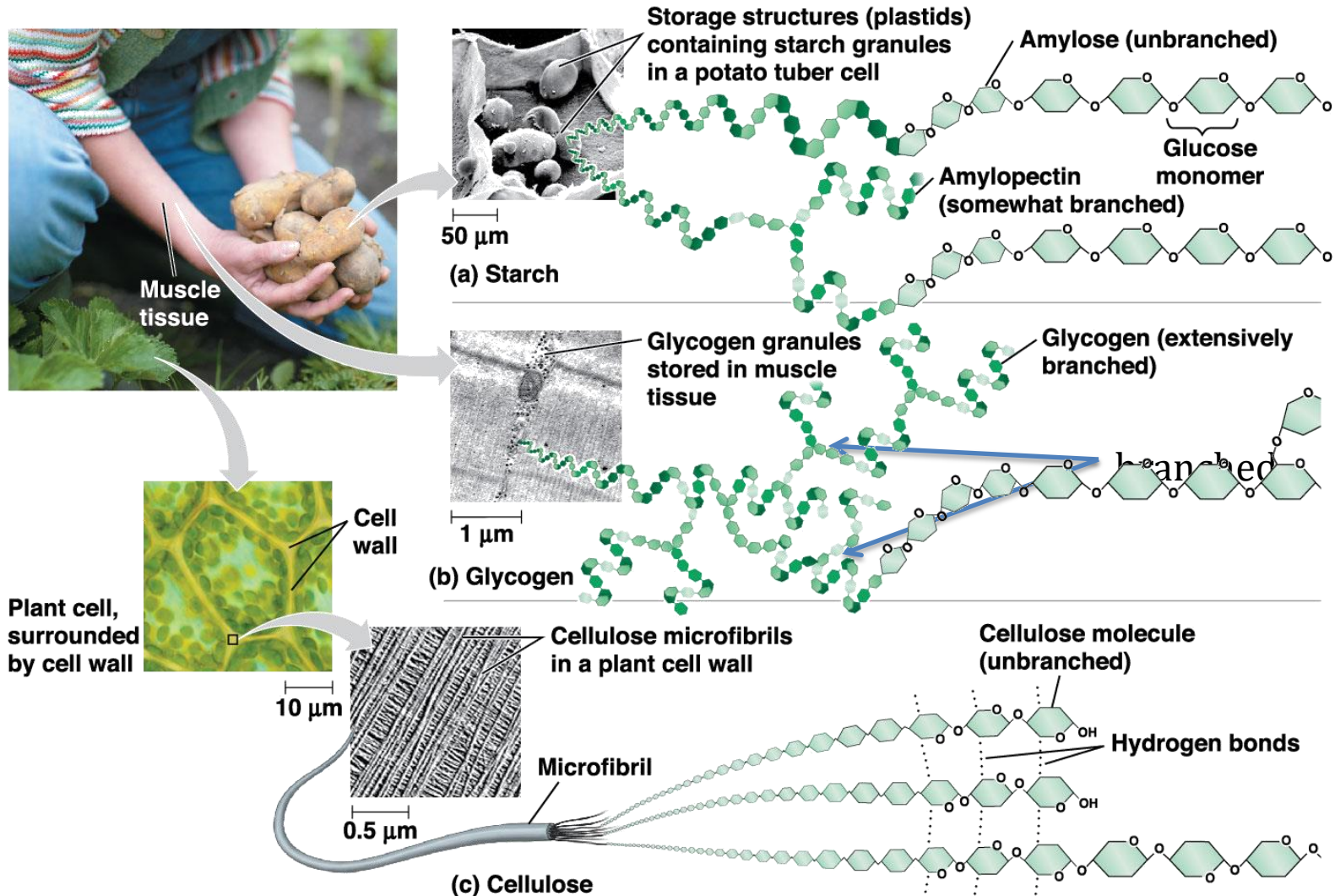
(b) Dehydration reaction in the synthesis of sucrose



Polysaccharides

- A **polysaccharide** consists of a **series** of connected **monosaccharides**. A polysaccharide is a **polymer**.
- Cells hydrolyze storage polysaccharides into sugars as needed.
- Common storage polysaccharides are **starch, glycogen, cellulose, and chitin**.

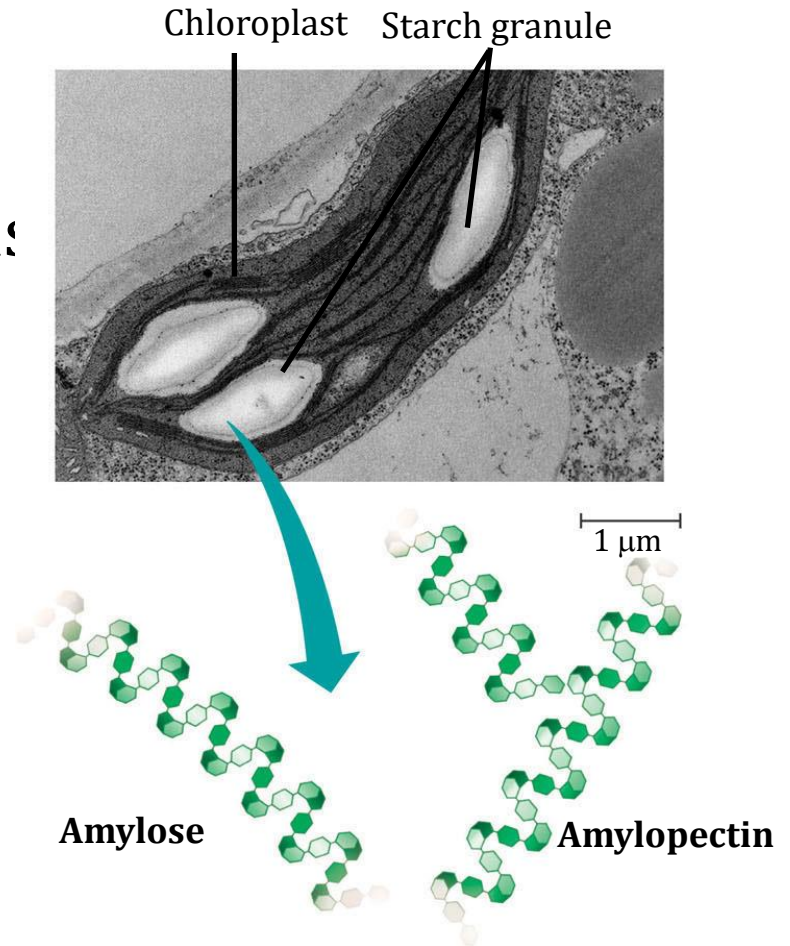
Polysaccharides of plants and animals



Storage polysaccharide in plants

- **Starch**

- α -glucose polymer that is molecule in plants

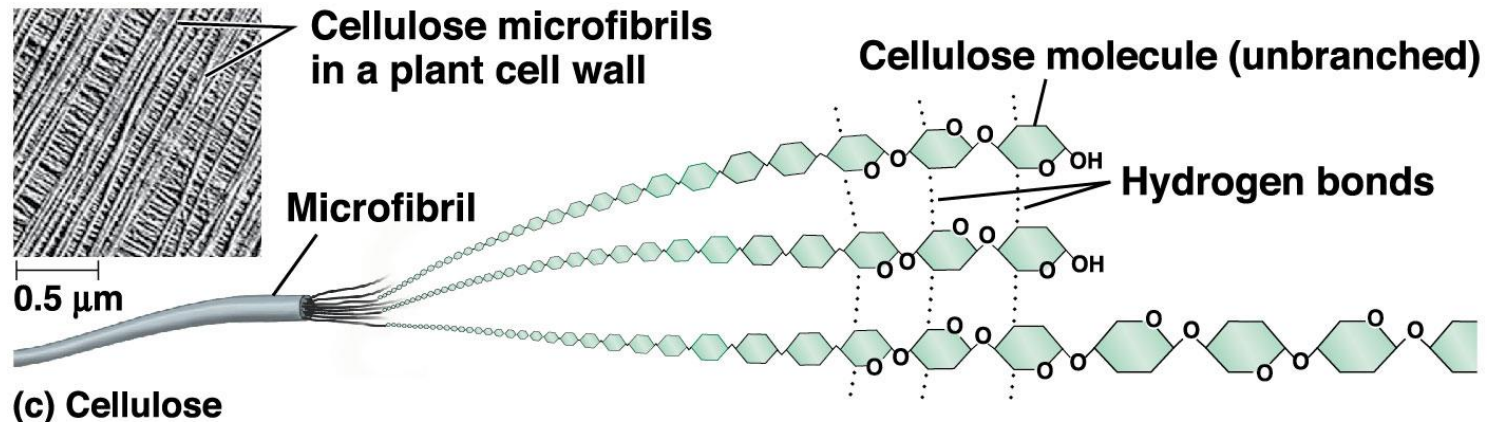


(a) Starch: a plant polysaccharide

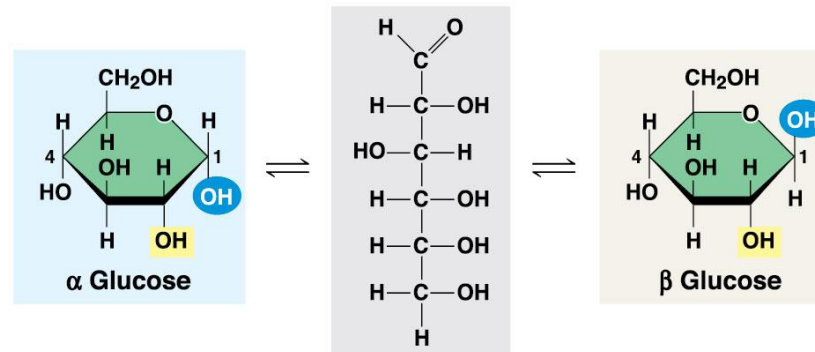
Structural polysaccharide in plants

- **Cellulose**

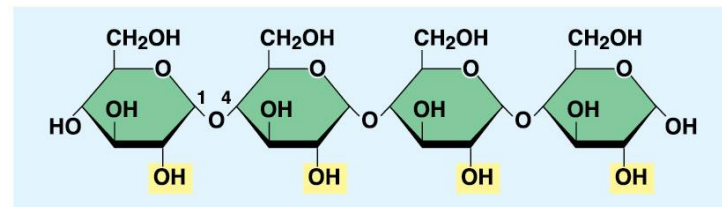
- polymer of β -glucose molecules.
- It serves as a structural molecule in the walls of plant cells.
- Cellulose is the major component in wood.



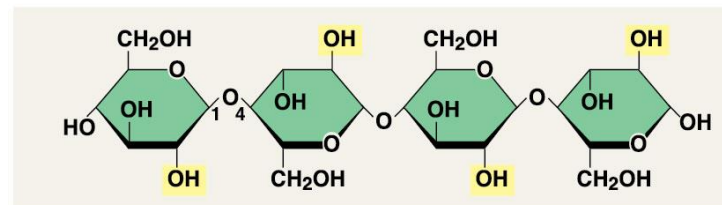
Starch and cellulose structure



(a) α and β glucose ring structures



(b) Starch: 1-4 linkage of α glucose monomers

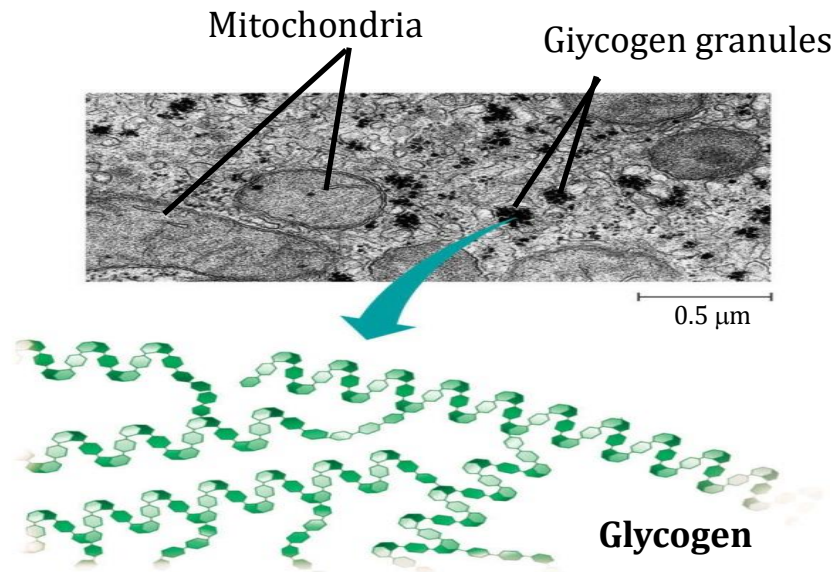


(c) Cellulose: 1-4 linkage of β glucose monomers

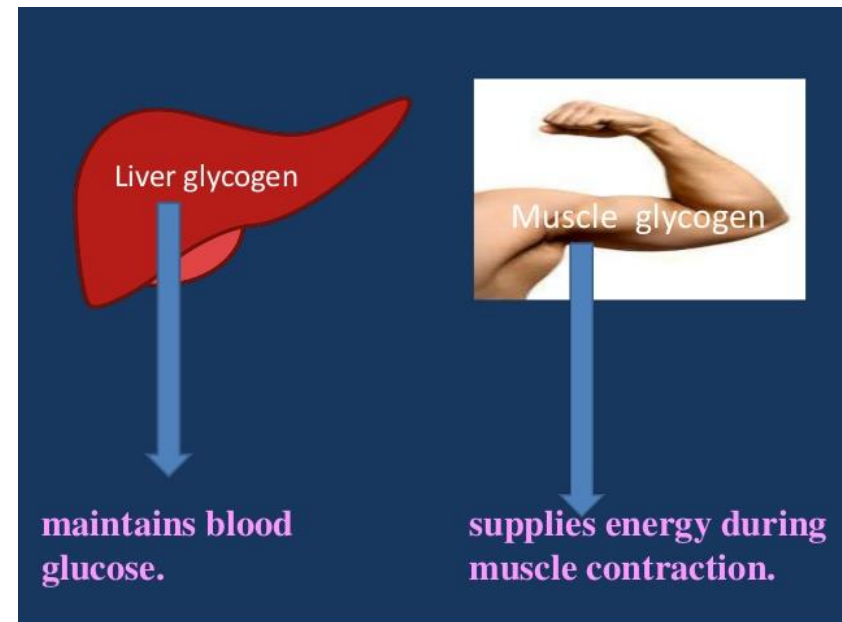
Storage polysaccharide in animals

- **Glycogen**

- α -glucose polymer
- more tightly branched than starch

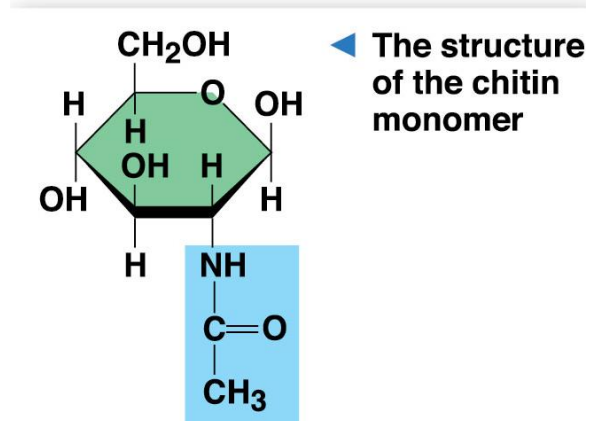


(b) Glycogen: an animal polysaccharide



Structural polysaccharide in animals

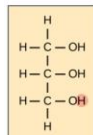
- **Chitin** is a polymer similar to cellulose, but each **β -glucose** molecule has a **nitrogen** containing group attached.
- Chitin serves as a structural molecule in the walls of fungus cells, exoskeletons of insects, and mollusks



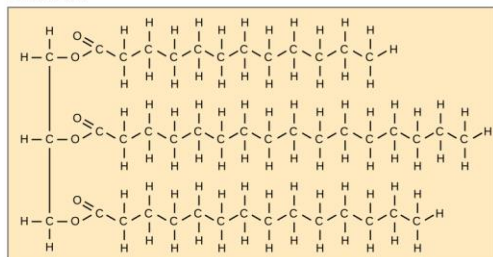
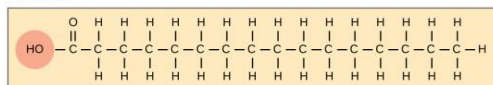
Chitin, embedded in proteins, forms the exoskeleton of arthropods.

Lipids

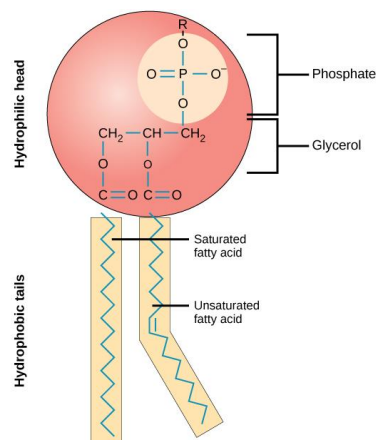
Triglyceride (fat)



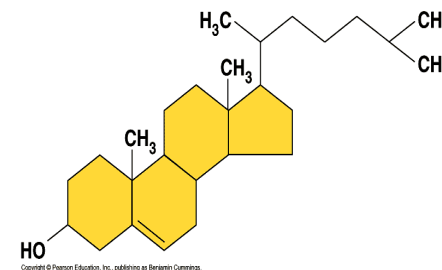
+



phospholipid

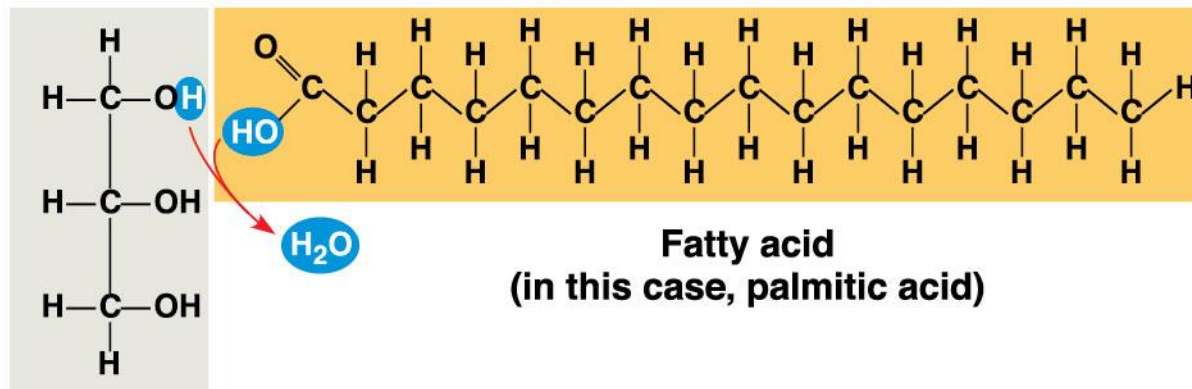


steroid



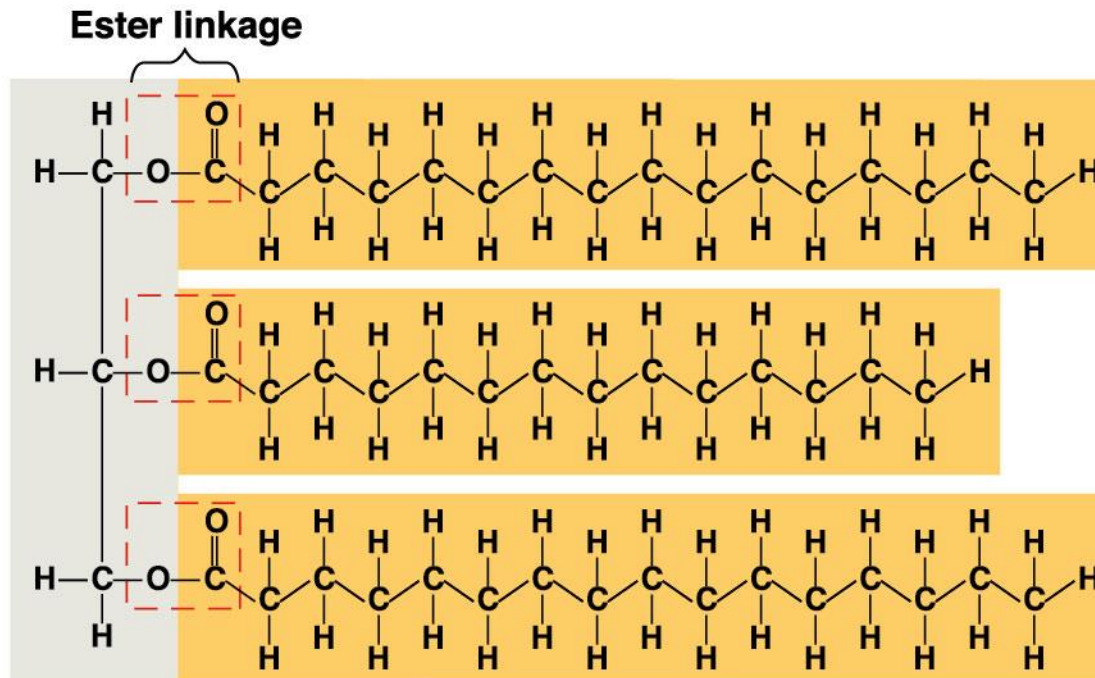
Fats/Triglyceride

- They consist of **3 fatty-acids** attached to a **glycerol** molecule.
- **Fatty acids** are **hydrocarbon chains** with a **carboxyl group** at one end of the chain
- Hydrophobic



Glycerol

(a) One of three dehydration reactions in the synthesis of a fat

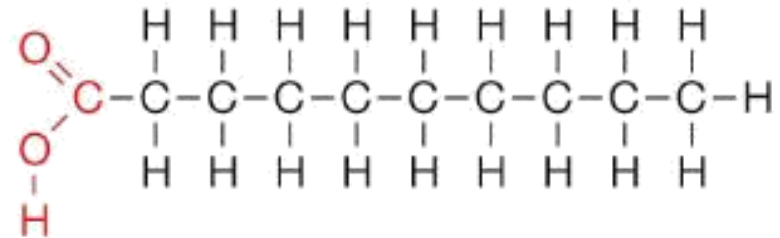


(b) Fat molecule (triacylglycerol)

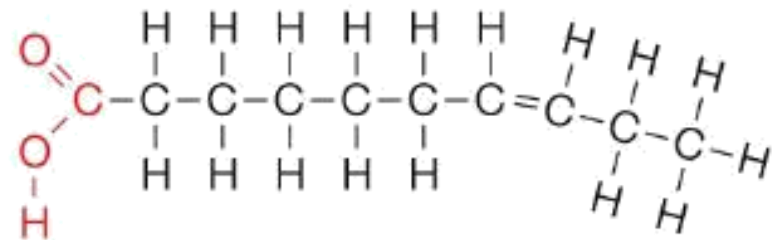
Hydrocarbon chains of fatty acids:

- HC chains of **Saturated fatty acids** have a
 - single covalent bond between each pair of carbon atoms, and
 - each carbon has 2 hydrogen bonded to it. The carbon is “saturated” with hydrogen
- HC chains of **Unsaturated fatty acids** have a
 - double covalent bond, and
 - each of the two carbons in this bond have only one hydrogen atom bonded to it.

Saturated

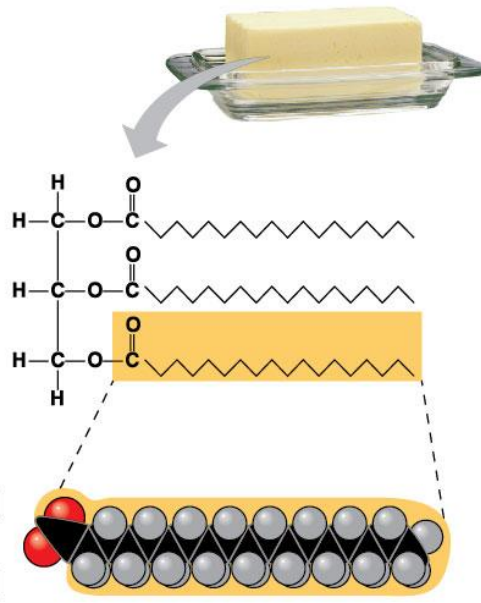


Unsaturated

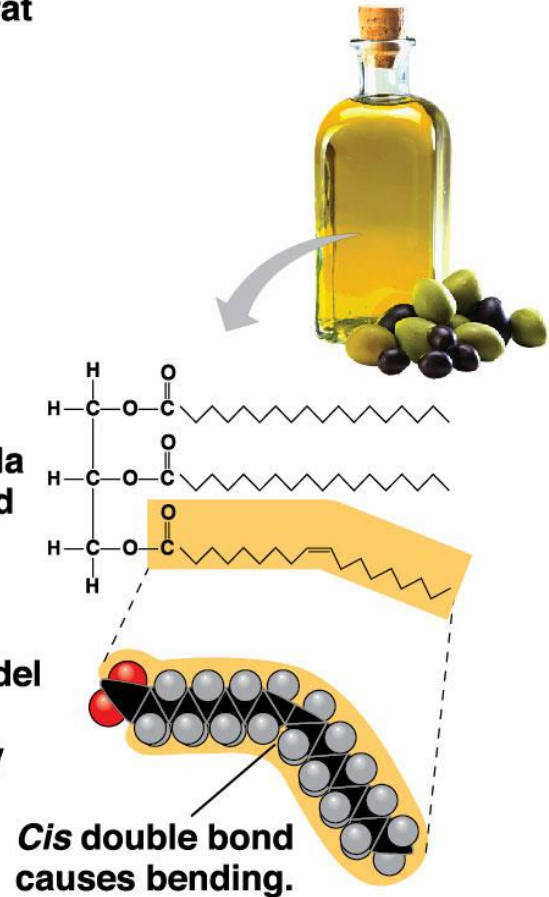
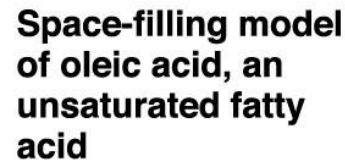


Saturated vs. Unsaturated

(a) Saturated fat

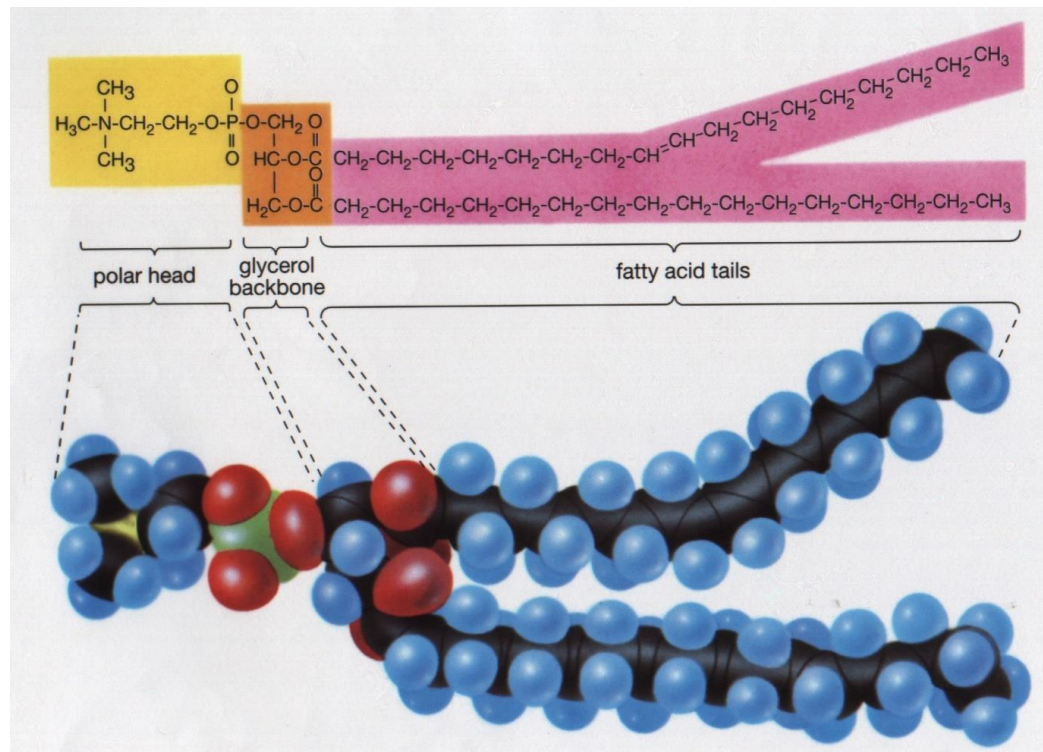


(b) Unsaturated fat



Phospholipids

- Have only two fatty acids
- Have a phosphate group instead of a third fatty acid



Phospholipids

- Phospholipid structure
 - Consists of a hydrophilic “head” and hydrophobic “tails”

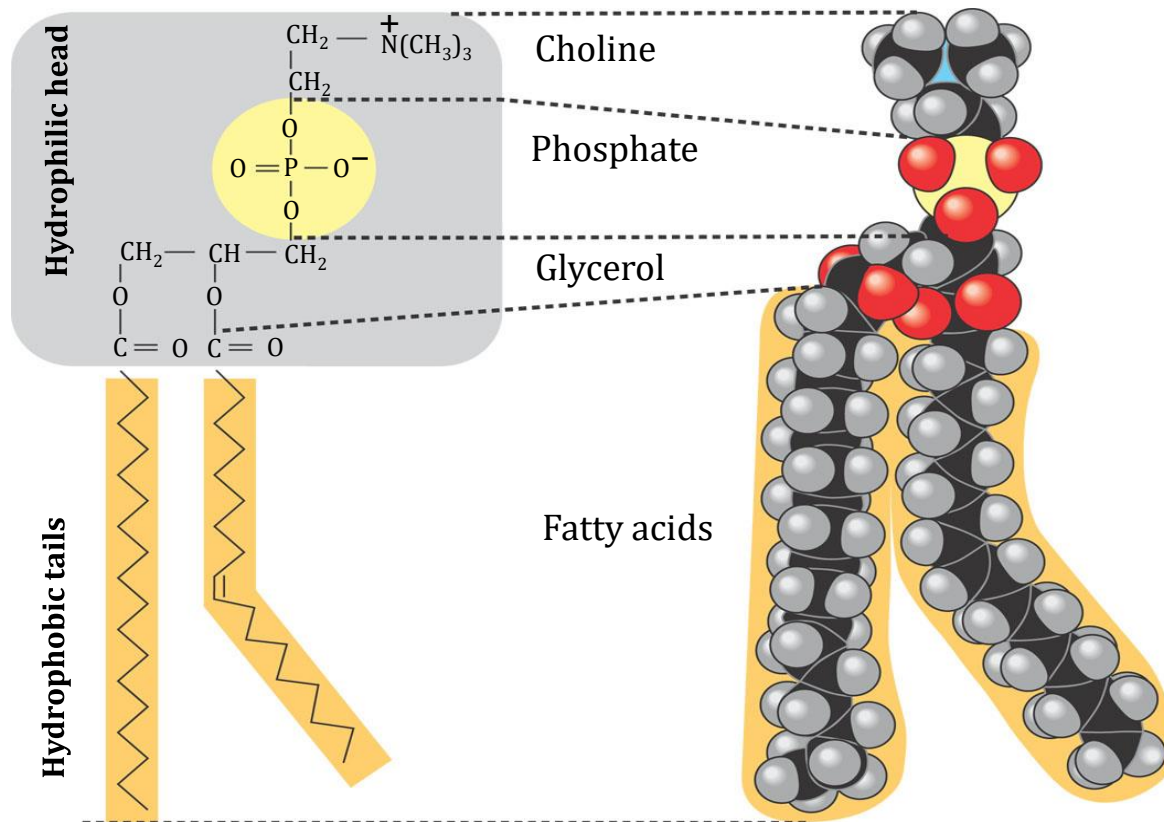


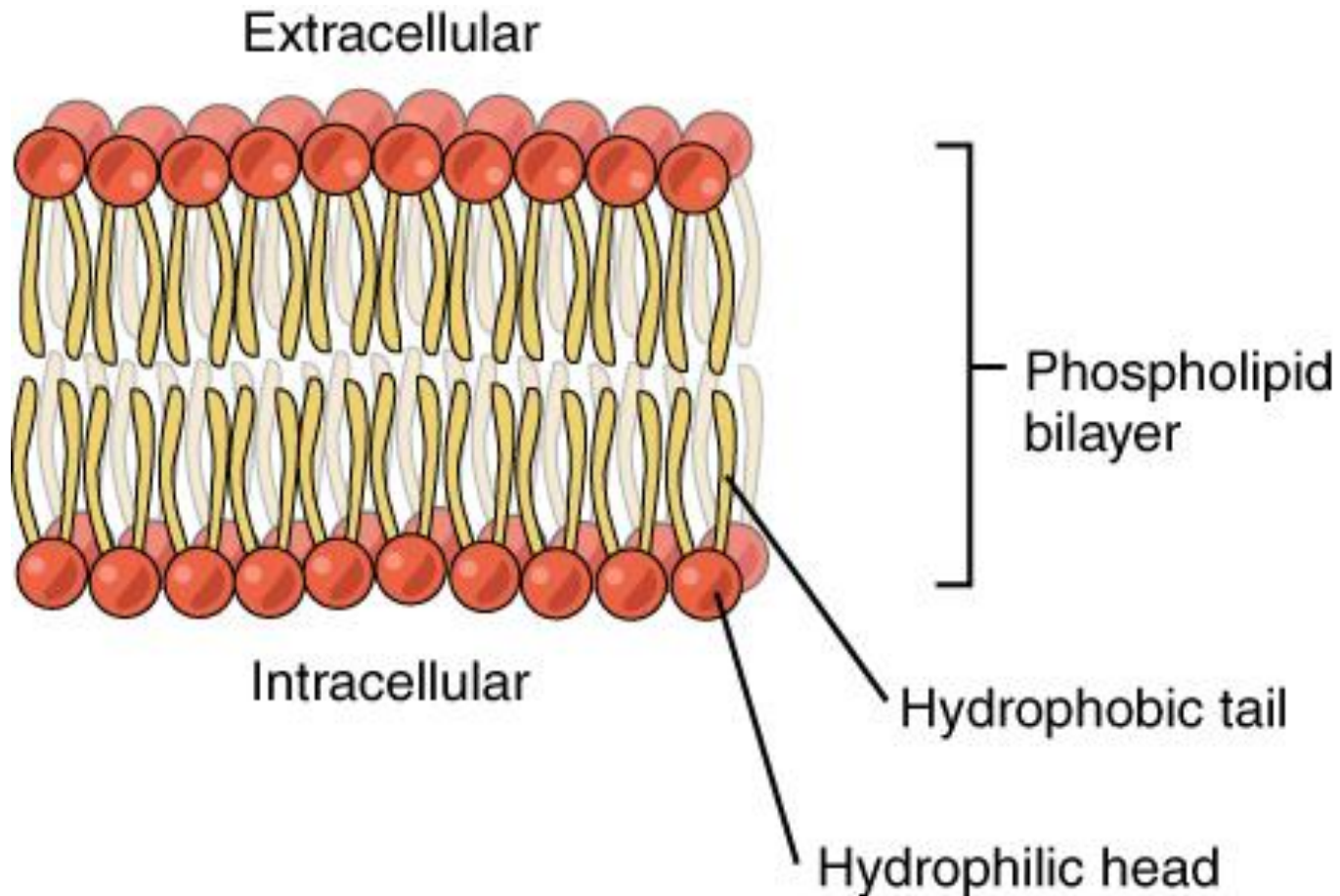
Figure 5.11

(a) Structural formula

(b) Space-filling model

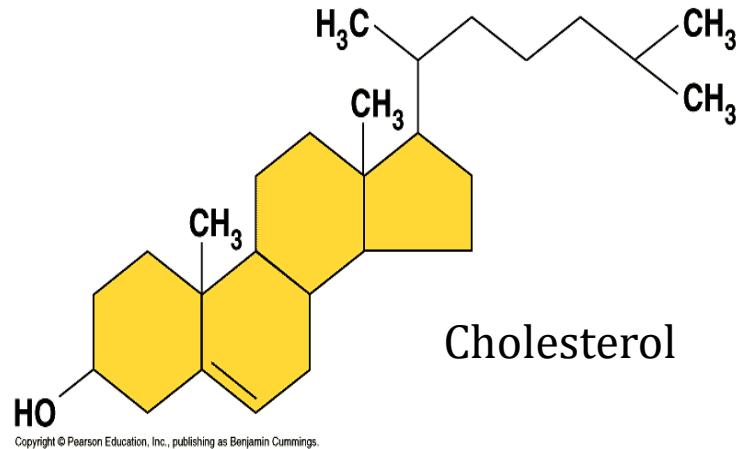
(c) Phospholipid symbol

Lipid bilayer



Steroids

- Steroids are characterized by a backbone of **four linked carbon** rings.
- Examples include cholesterol (a component of cell membranes), hormones, testosterone and estrogen.

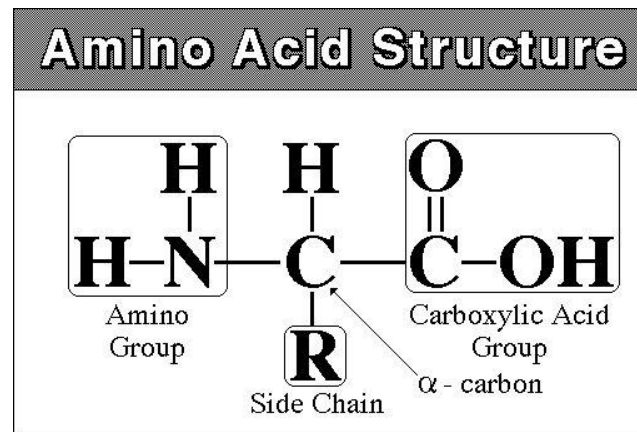


Proteins: functions

- Proteins can be grouped according to their functions:
 - Structural proteins: keratin, hair, horns, collagen, connective tissue, spider silk
 - Storage proteins: casein in milk & ovalbumin in egg whites
 - Transport proteins: found on cell membranes that transport materials into and out of cell, hemoglobin
 - Defensive proteins: antibodies to fight infection
 - Enzymes: regulate the rate of chemical reactions

Proteins: building blocks

- Amino acids
 - Are organic molecules possessing both carboxyl and amino groups
 - Differ in their properties due to differing side chains, called R groups

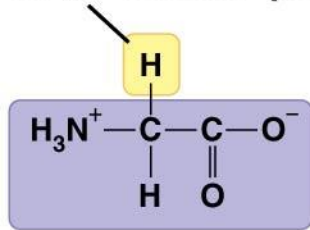


The 20 Amino Acids:

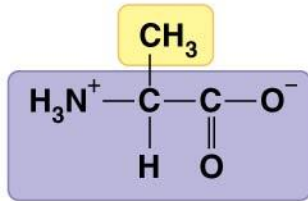
Nonpolar side chains

Nonpolar side chains; hydrophobic

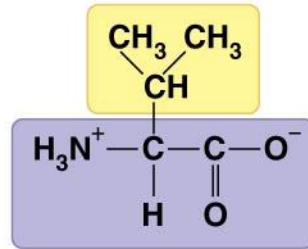
Side chain (R group)



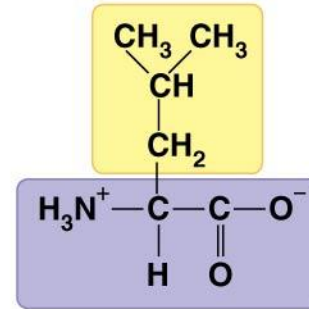
Glycine
(Gly or G)



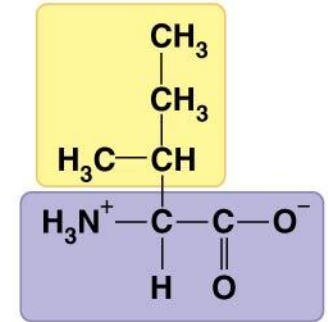
Alanine
(Ala or A)



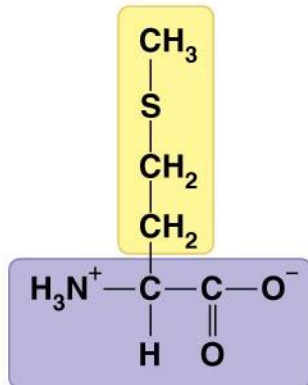
Valine
(Val or V)



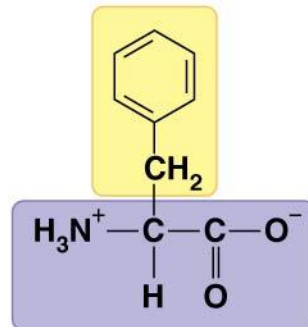
Leucine
(Leu or L)



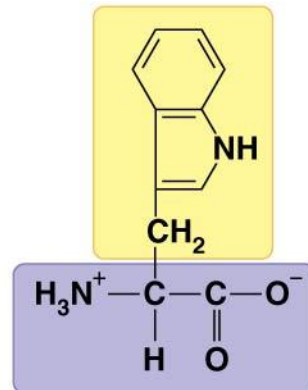
Isoleucine
(Ile or I)



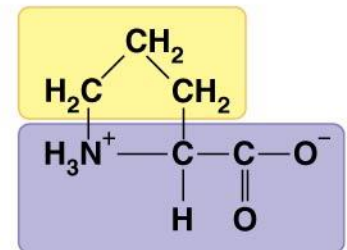
Methionine
(Met or M)



Phenylalanine
(Phe or F)



Tryptophan
(Trp or W)

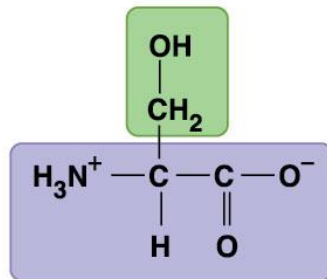


Proline
(Pro or P)

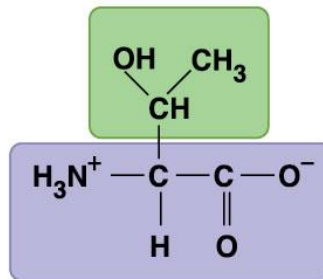
The 20 Amino Acids:

Polar side chains

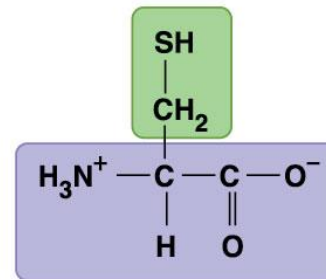
Polar side chains; hydrophilic



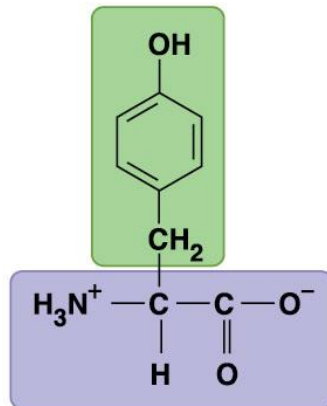
Serine
(Ser or S)



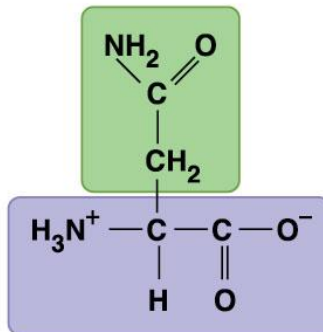
Threonine
(Thr or T)



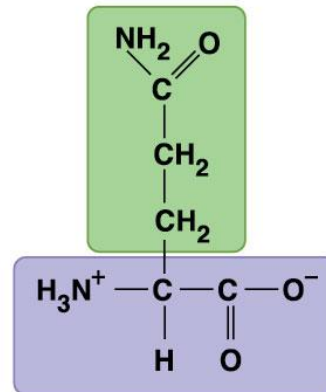
Cysteine
(Cys or C)



Tyrosine
(Tyr or Y)



Asparagine
(Asn or N)



Glutamine
(Gln or Q)

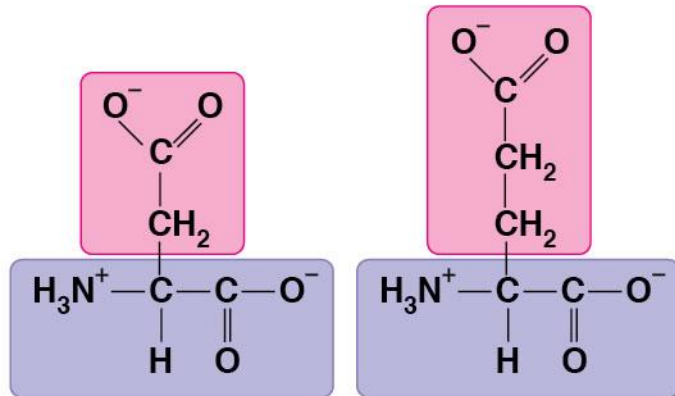
The 20 Amino Acids:

Electrically charged side chains

Electrically charged side chains; hydrophilic

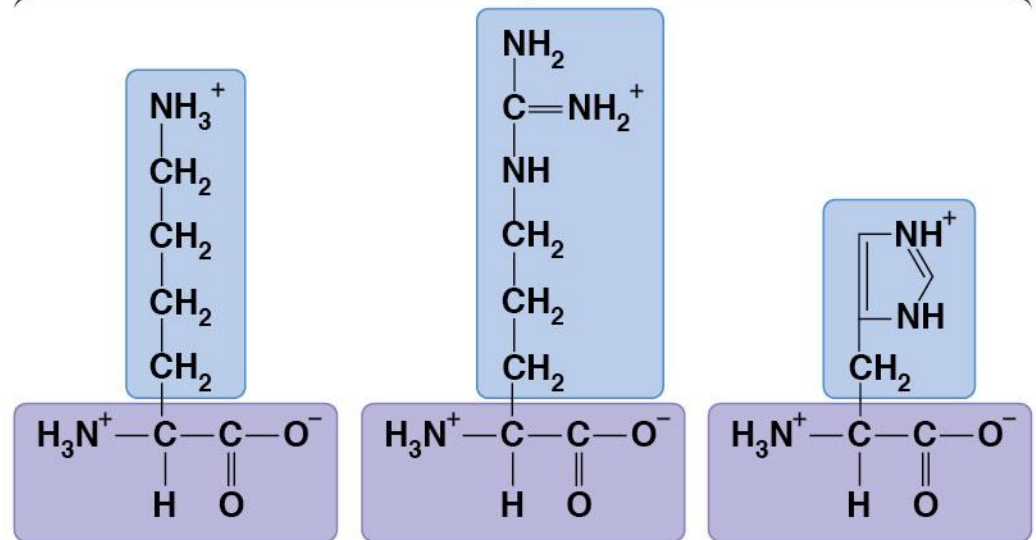
Basic (positively charged)

Acidic (negatively charged)



Aspartic acid
(Asp or D)

Glutamic acid
(Glu or E)



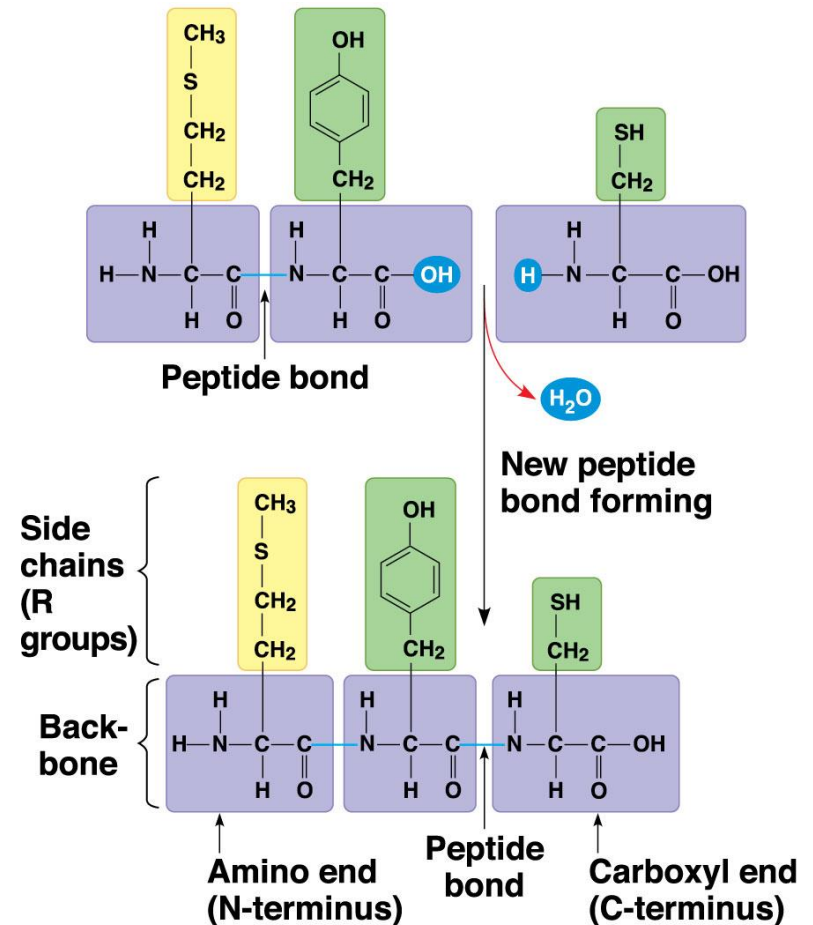
Lysine
(Lys or K)

Arginine
(Arg or R)

Histidine
(His or H)

Amino acids to polypeptide

- Dehydration reaction
(-H₂O)
- Linking **carboxyl group** of one amino acid with **amino group** of the next
- **Peptide bond** connects polypeptide backbone



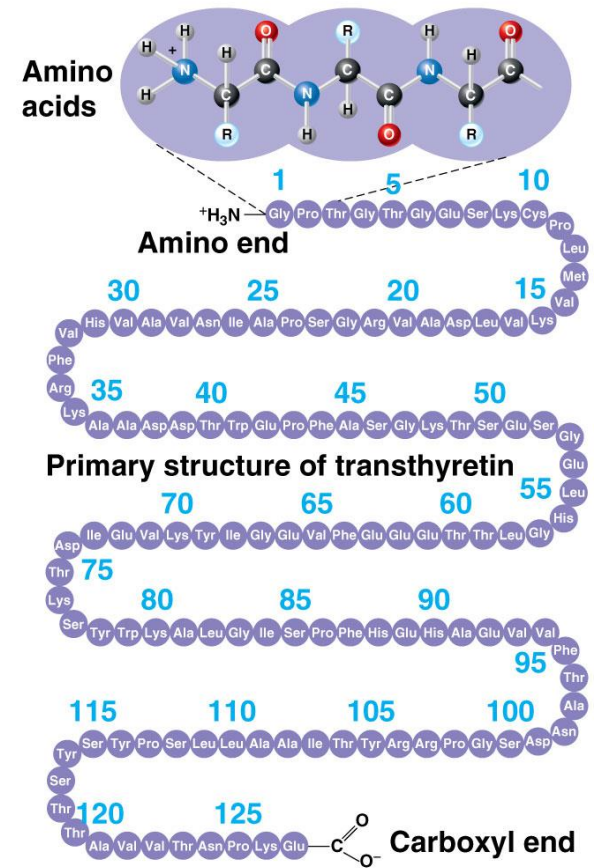
Proteins

- For a protein to function properly, it has to be folded into a specific structure. Protein folding has four levels of “folding”
 - Primary
 - Secondary
 - Tertiary
 - Quarternary

Primary Structure

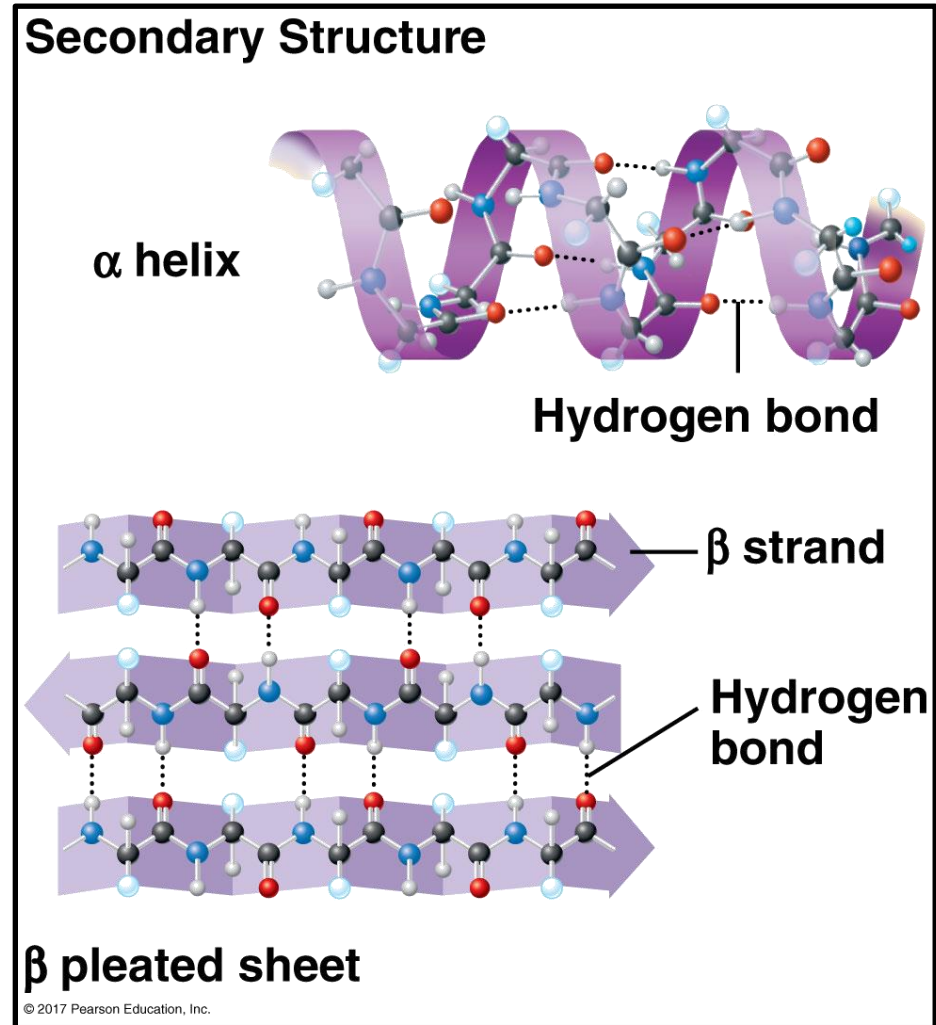
- The primary structure of a protein describes the **linear order of amino acids**.
- Peptide bond

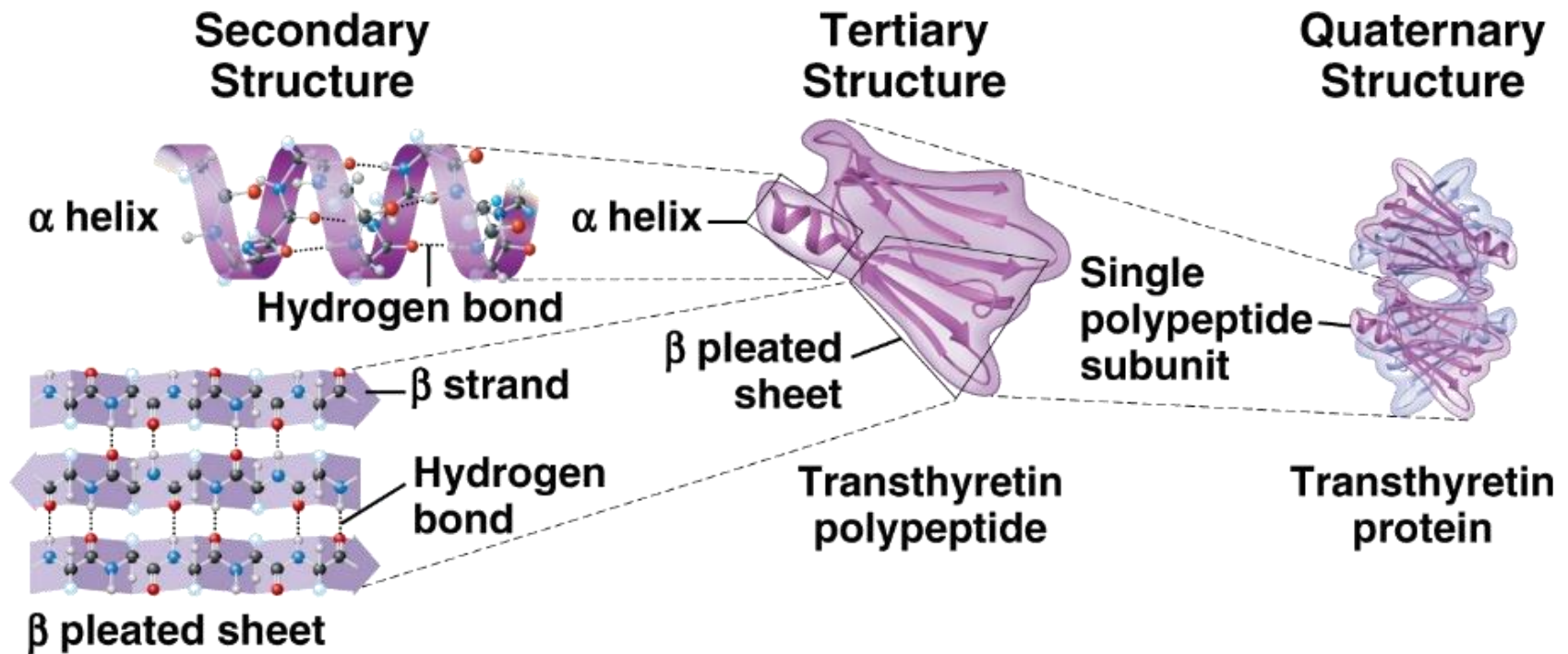
Primary Structure



Secondary Structure

- Secondary structure of a protein is a **3D shape** that results from hydrogen bonding
 - between the **amino** and **carboxyl groups** of adjacent amino acids (polypeptide backbone)
 - Individually, the H-bonds are weak
 - but repetition over a long region can support a particular shape
- The bonding can produce a **spiral** structure called **alpha helix**
- **Or a folded** structure called **beta pleated sheet**
- Proteins whose shape is dominated by these 2 patterns, form fibrous proteins

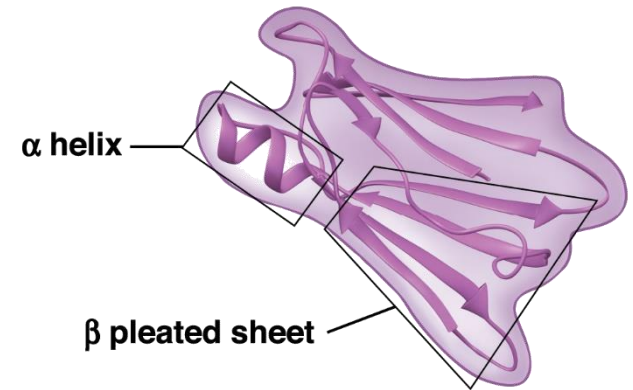




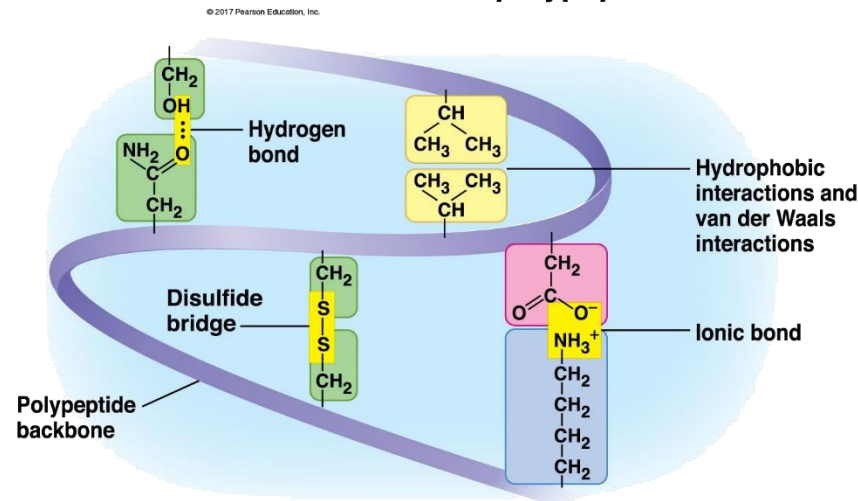
Tertiary Structure

- Tertiary structure of a protein includes additional 3D shaping. The following factors contribute to tertiary “folding”
 - Hydrogen bond between R groups of amino acids
 - Ionic bonding between R groups of amino acids
 - Hydrophobic interaction
 - Disulfide bonds, when the sulfur atoms in two cysteine amino acids bond → “disulfide bridge”

Tertiary Structure



Transthyretin polypeptide

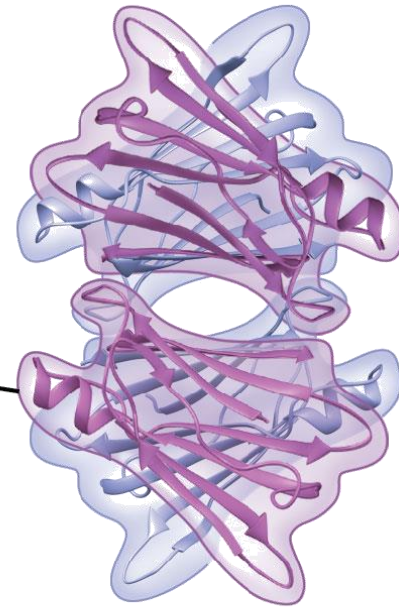


Quaternary Structure

- Quaternary structure describes a protein that is **assembled from two or more separate polypeptides**.
 - E.g. Transthyretin is made up of 4 polypeptides
- some proteins only

Quaternary Structure

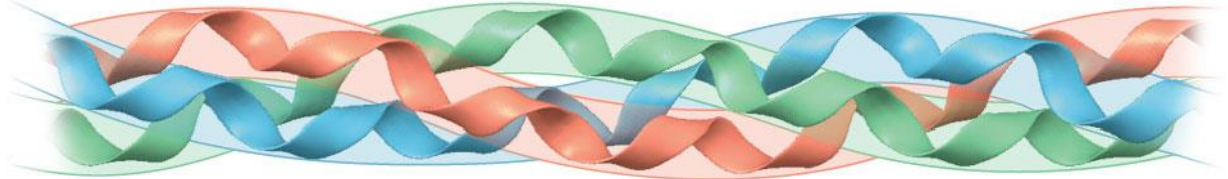
Single polypeptide subunit



Transthyretin protein

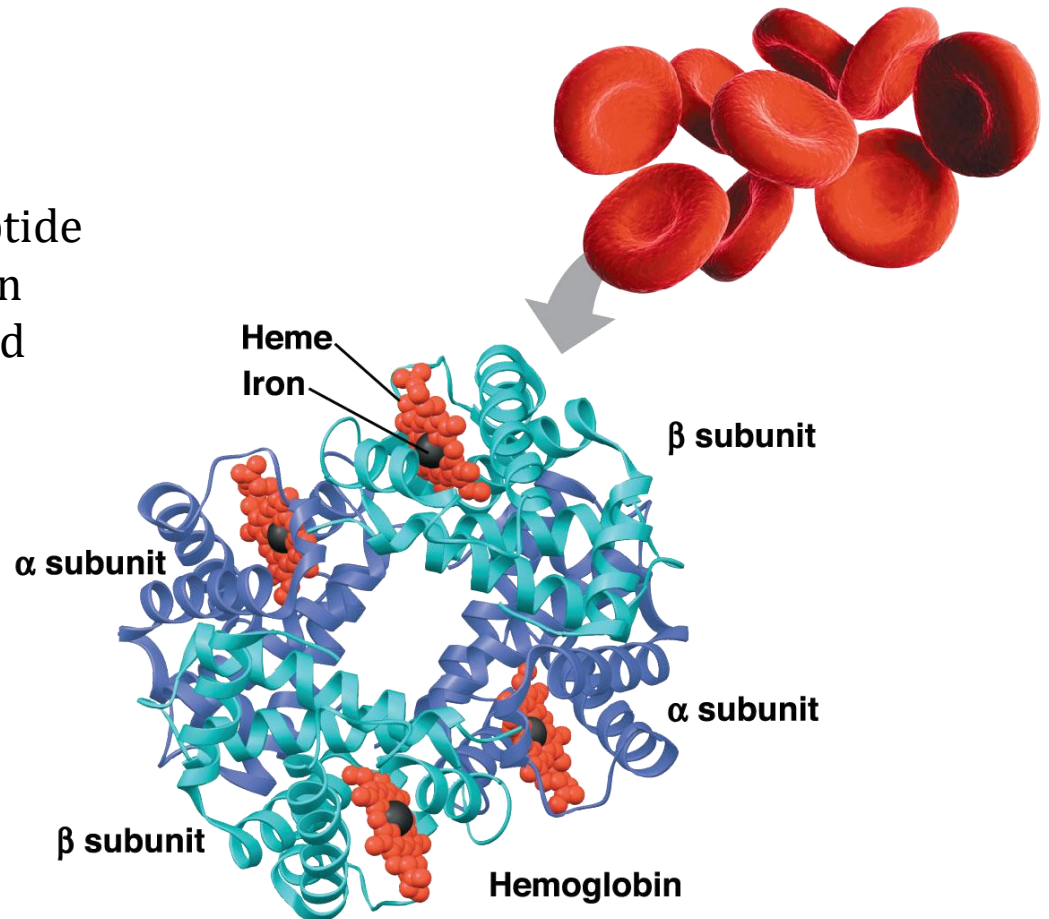
Collagen has 3 helical polypeptides intertwined into a larger triple helix

Collagen



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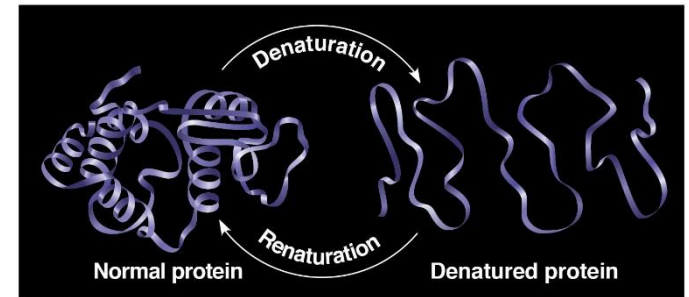
Hemoglobin consists of 4 polypeptide subunits held together by hydrogen bonding, R-groups interactions, and disulfide bonds



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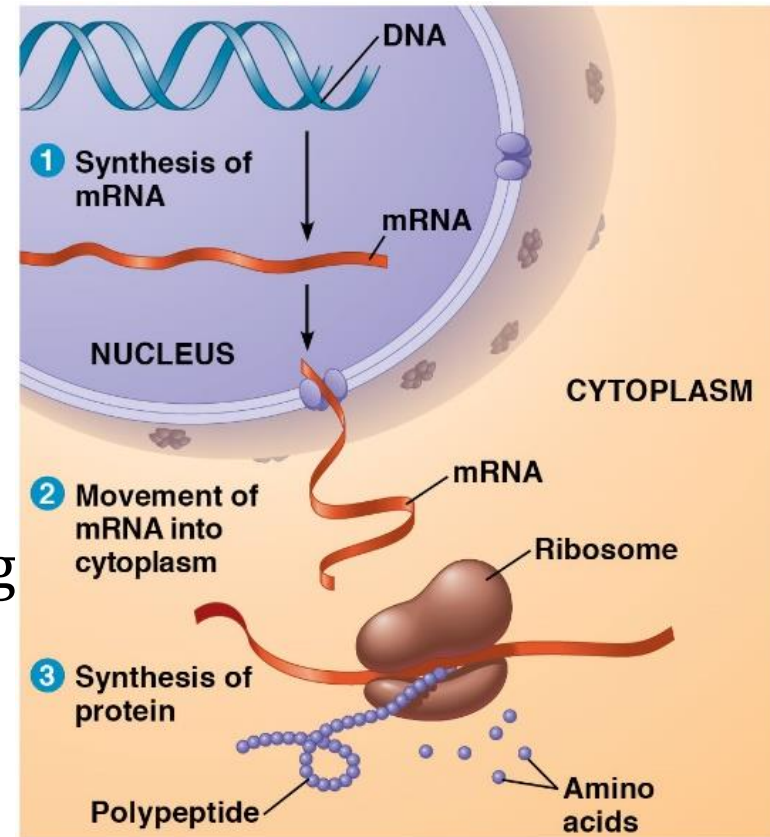
What determines protein structure?

- Physical and chemical conditions of proteins' environment
 - **Denatured** proteins no longer work in their unfolded condition
 - Proteins may be denatured by changes in
 - pH
 - salt concentration
 - Temperature
 - Environment



Nucleic Acids: Role

- 2 types:
 - DNA (**D**eoxyribo**N**ucleic **A**cid)
 - Inheritable genetic material
 - Can direct synthesis of mRNA
 - RNA (**R**ibo**N**ucleic **A**cid)
 - Conveys genetic info for building proteins
 - controls protein synthesis

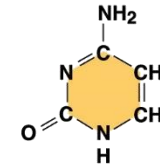


Nucleic acids: structure

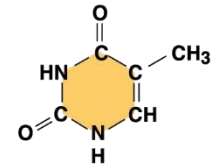
- Nucleic acids are made up of:
 - **nitrogenous bases**
 - Cytosine, Thymine and Uracil and are “**Pyrimidines**” single-ring bases
 - Adenine and Guanine are **Purines** double-ring bases
 - **5-Carbon sugar**
 - Deoxyribose
 - Ribose
- Nitrogenous base + sugar = nucleoside

NITROGENOUS BASES

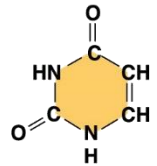
Pyrimidines



Cytosine (C)

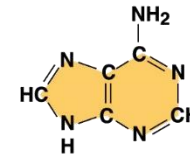


Thymine
(T, in DNA)

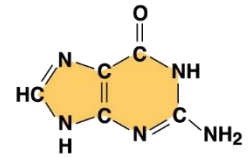


Uracil
(U, in RNA)

Purines



Adenine (A)

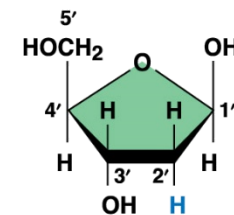


Guanine (G)

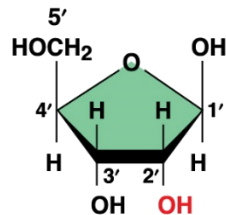
(c) Nucleoside components

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SUGARS



Deoxyribose
(in DNA)

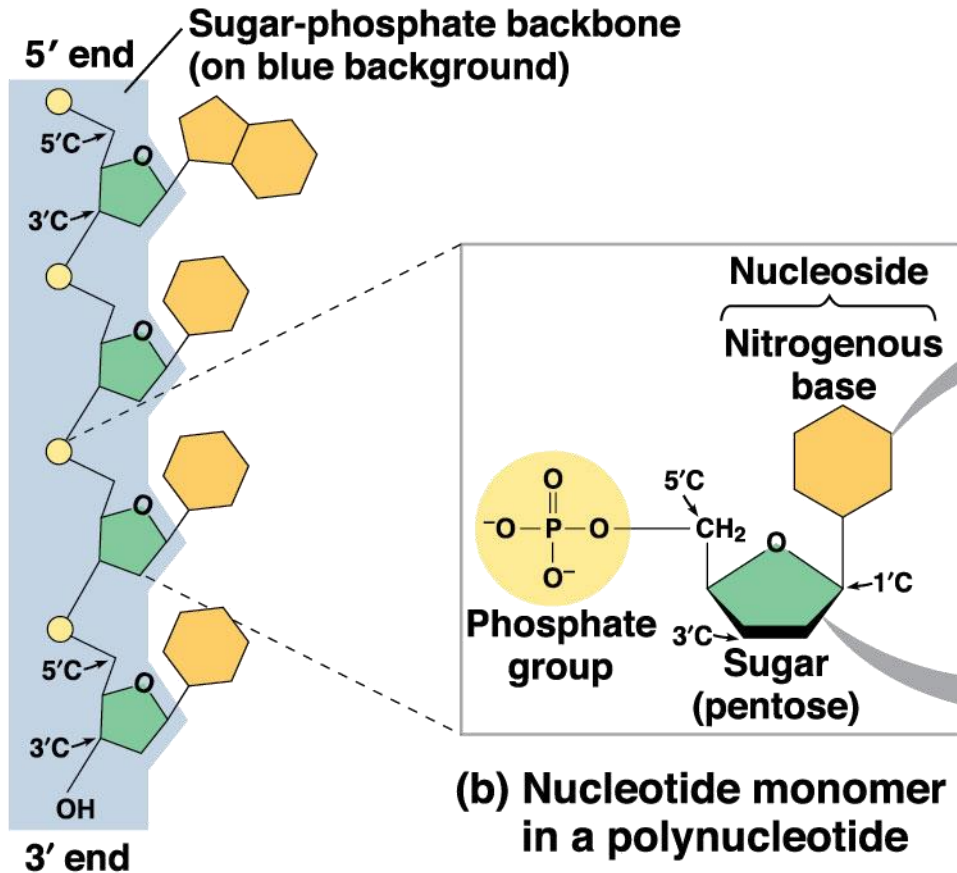


Ribose
(in RNA)

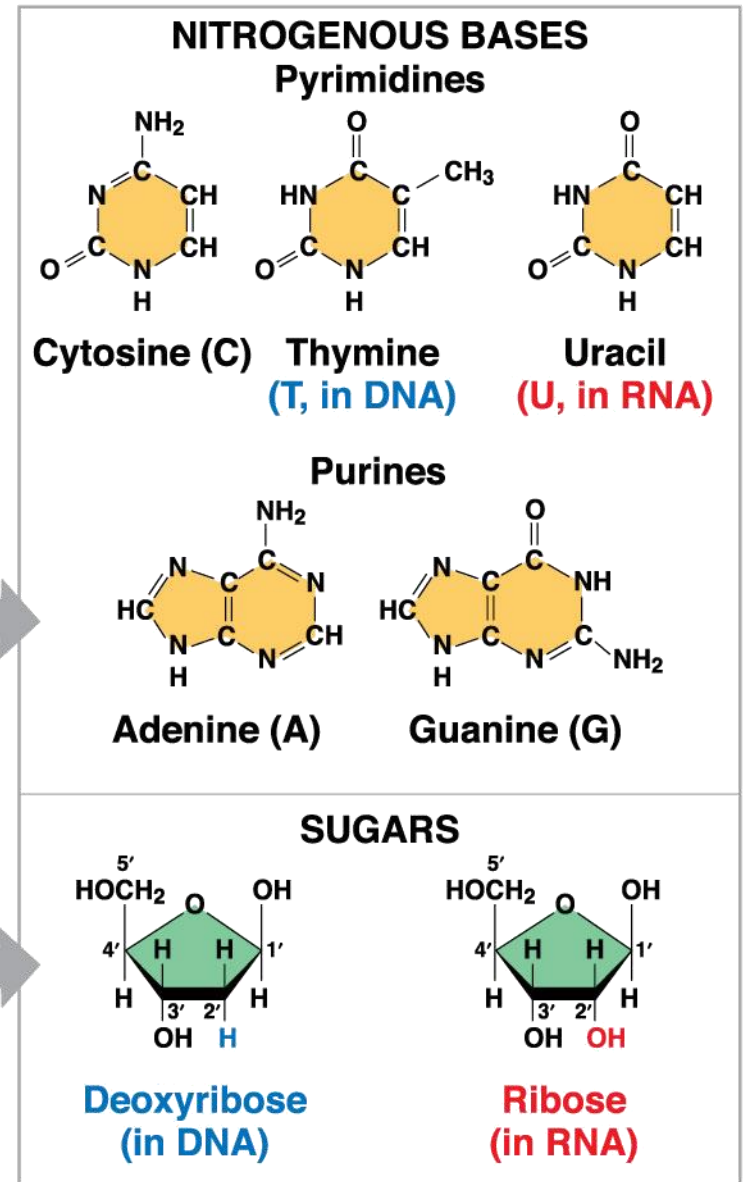
(c) Nucleoside components

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- Nucleoside + phosphate group = **nucleotide**



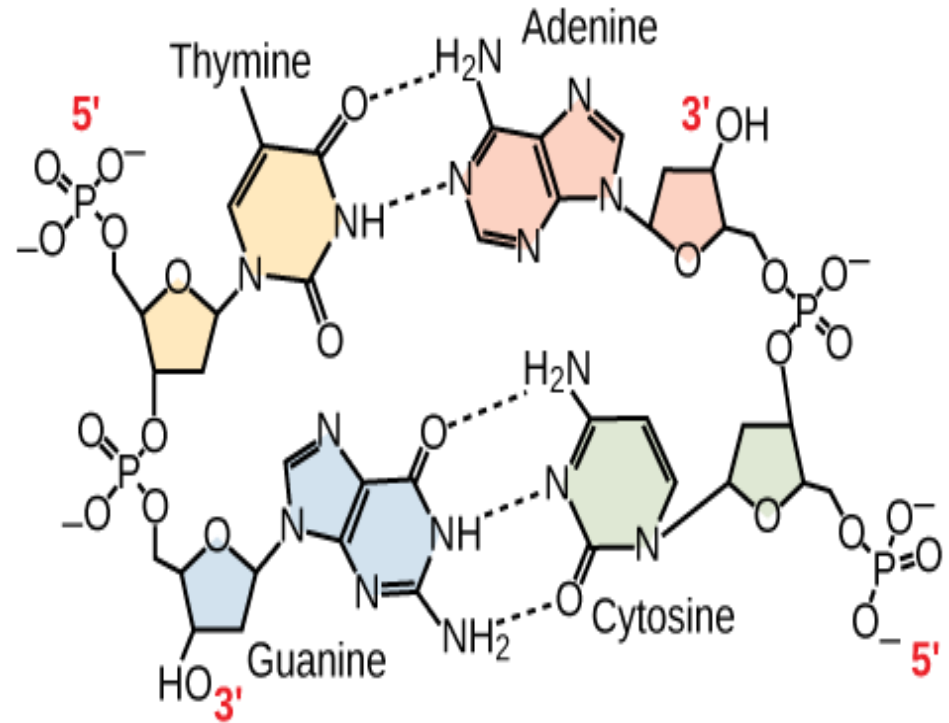
(a) Polynucleotide, or nucleic acid



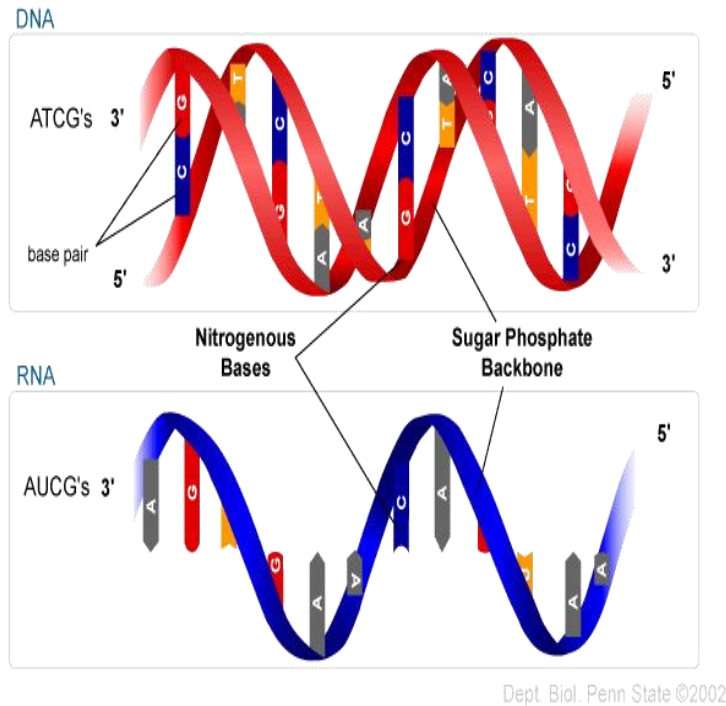
(c) Nucleoside components

Nucleic Acids: Structure of DNA and RNA

- The **two strands** of DNA are **antiparallel**: they run in opposite directions.
- Form a **double helix**
- One is arranged in the 5'→3' direction the other in the 3'→5' direction



RNA



- RNA differs from DNA in the following ways:
 - The sugar in RNA is ribose, not deoxyribose
 - The thymine does not occur in RNA, it is replaced with uracil
 - RNA is a **single-stranded molecule** and does not form a double helix as DNA does.

Features of DNA and RNA

	DNA	RNA
Function	Carries genetic information	Involved in protein synthesis
Location	Remains in the nucleus	Leaves the nucleus
Structure	Double helix	Usually single-stranded
Sugar	Deoxyribose	Ribose
Pyrimidines	Cytosine, thymine	Cytosine, uracil
Purines	Adenine, guanine	Adenine, guanine

Summary I

- **Matter** consists of chemical **elements** in pure form and in **combinations** called **compounds**
- An **element's** properties depend on the structure of its **atoms**
- The formation and function of molecules depend on **chemical bonding** between atoms
- **Polar covalent bonds** in water molecules result in **hydrogen bonding**
- **4 emergent properties of water** contributes to Earth's suitability for life
- **Acidic and basic conditions affect living organisms**

Summary II

- Carbon atoms can form diverse molecules by bonding to four other atoms
- A few **chemical groups** are key to molecular function
- **Macromolecules** are polymers made from monomers
- **Carbohydrate** serve as fuel and building material
- **Lipids** are a diverse group of lipid macromolecules
- **Proteins** include a diversity of structures, resulting in a wide range of functions
- **Nucleic acids** store, transmit, and help express hereditary information